

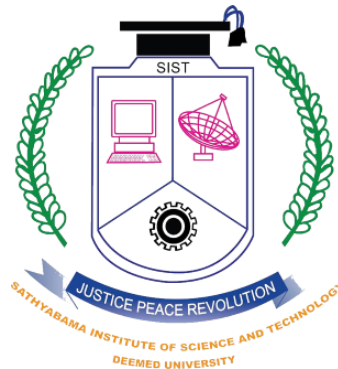
AYUSH VIRTUAL HERBAL GARDEN

Submitted in partial fulfillment of the requirements for the award of
Bachelor of Engineering degree in Computer Science and
Engineering in Data Science

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING

SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)**

CATEGORY - 1 UNIVERSITY BY UGC

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APRIL - 2026



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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **KOLLA HARSHITHA (42733043), PARESIREDDI GARI LIKHITHA (42733045)**, who carried out the Project entitled "**Ayush Virtual Herbal Garden**" under my supervision from November 2025 to April 2026.

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I, **KOLLA HARSHITHA** (Reg. No- 42733043), hereby declare that the Project Report entitled "**AYUSH VIRTUAL HERBAL GARDEN**" done by me under the guidance of **Dr. V. SUBAPRIYA, M.E., Ph.D.**, is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in **Computer Science and Engineering with Data Science**.

DATE: 16/04/2026

PLACE: Chennai

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ACKNOWLEDGEMENT

I am pleased to acknowledge my sincere thanks to **Board of Management of SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY** for their kind encouragement in doing this project and for completing it successfully. I am grateful to them.

I convey my thanks to **Dr. L. LAKSHMANAN, M.E., Ph.D., DEAN**, School of Computing and **Dr. SENDURU SRINIVASULU, M.Tech., Ph.D., HEAD**, Department of Computer Science and Engineering for providing me necessary support and details at the right time during the progressive reviews.

I would like to express my sincere and deep sense of gratitude to my Project Guide **Dr. V. SUBAPRIYA, M.E., Ph.D.**, for her valuable guidance, suggestions and constant encouragement paved way for the successful completion of my project work.

I wish to express my thanks to all Teaching and Non-teaching staff members of the **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING WITH DATA SCIENCE** who were helpful in many ways for the completion of the project.

ABSTRACT

The AYUSH Virtual Herbal Garden offers an innovative and practical solution for identifying Ayurvedic medicinal plants through AI-powered technology, seamlessly integrating traditional knowledge into modern healthcare practices. The system leverages a dataset of approximately 2,000 images spanning 30 species of medicinal leaves, utilizing Convolutional Neural Networks (CNN) to deliver high classification accuracy. Enhanced by Transfer Learning, the plant identification process ensures efficiency and precision, making it a reliable tool for users. The Flask-based web interface serves as the backbone of the system, providing a user-friendly platform that delivers comprehensive plant information—including scientific names, medicinal uses, and geographical details—alongside advanced interactive features. Key functionalities include a Disease and Remedy Module, which allows users to input diseases and receive tailored Ayurvedic remedy suggestions based on a predefined knowledge base of medicinal plants. The Prediction Module employs AI to identify plant species from uploaded images, offering detailed scientific and geographical insights. Extending its scope, the 'Herbal Teas Module enriches the system by providing users with herbal tea recipes and their associated health benefits, drawing from the medicinal properties of identified plants.

Keywords: AYUSH, Medicinal Plant Identification, Transfer Learning, Ayurvedic Remedies, Traditional Medicine.

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM STATEMENT

The AYUSH sector, which encompasses Ayurveda, Yoga, Unani, Siddha, and Homeopathy, relies heavily on medicinal plants for traditional healing practices. However, access to physical herbal gardens is often restricted due to geographical limitations, making it difficult for individuals to explore and learn about these valuable plants. Additionally, identifying medicinal plants correctly requires specialized botanical knowledge, which is not commonly available to students, general practitioners, or enthusiasts. This lack of accessibility and expertise results in misidentification, which can lead to incorrect usage, potential health hazards, and a decline in trust in traditional remedies.

Existing solutions, such as books and online databases, provide theoretical knowledge but fail to offer an interactive and immersive learning experience. Many plant identification tools available today cater to general botanical studies but do not specifically focus on AYUSH medicinal plants, making them less effective for users who require targeted information. There is a pressing need for an innovative digital solution that simplifies plant identification, provides authentic AYUSH-based medicinal knowledge, and engages users in an interactive and immersive learning environment. This platform should bridge the gap between traditional knowledge and modern technology, ensuring the preservation and widespread accessibility of herbal wisdom.

1.2 OBJECTIVE

The Virtual Herbal Garden aims to bridge the knowledge gap in AYUSH-based traditional medicine by providing an interactive, educational, and immersive digital experience. By integrating AI-powered plant identification, users can upload images of plants and receive accurate identifications along with their medicinal properties and traditional uses.

Additionally, the system will provide personalized herbal medicine recommendations based on user's health concerns, ensuring that natural remedies are easily accessible to those interested in holistic healing.

To further promote wellness and preventive healthcare, the platform will suggest yoga postures that align with Ayurvedic treatment principles. By incorporating voice-based search, text-to-speech conversion, and multi-language support, the Virtual Herbal Garden will enhance accessibility, making it inclusive for users of different backgrounds and abilities. Through this fusion of technology and traditional wisdom, the project will encourage a deeper understanding of natural healing practices and sustainable healthcare solutions.

1.3 METHODOLOGY OVERVIEW

The proposed system follows an Agile methodology, structured into six 2-week sprints over a 12-week timeline to ensure systematic and incremental development. Agile development allows flexibility, continuous user feedback, and iterative improvements, making it the ideal choice for this project. Each sprint was designed to deliver a functional component of the system, allowing for adjustments based on feedback received in sprint reviews.

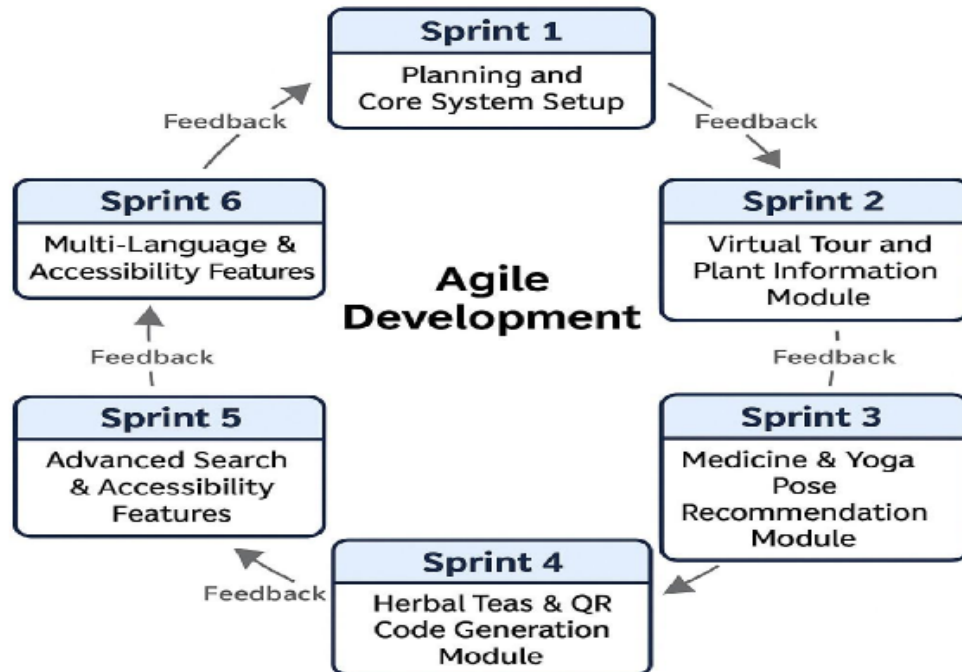


Fig 1.1 Agile Methodology

Sprint 1: Planning and Core System Setup

The project began with defining the scope, objectives, and features to be implemented. The goal was to establish a clear roadmap for development while ensuring the system effectively promotes AYUSH practices. A key focus was designing an intuitive user interface that encourages engagement and enhances awareness of Ayurveda, Yoga, and traditional medicine. Flask was chosen as the backend framework due to its lightweight nature and flexibility, allowing for easy integration of AI models. AI-driven medicinal plant identification was implemented using EfficientNet and ResNet models, trained on a dataset of 2,000 images representing 30 AYUSH medicinal plant species.

Core functionalities were validated to ensure the AI model could accurately classify medicinal plants based on image recognition. User input mechanisms were established to allow feedback collection, forming the basis for iterative improvements in future sprints.

Sprint 2: Plant Information Module

High-resolution images of plants were integrated along with detailed information, including their scientific names, medicinal benefits, and traditional applications. Each plant was accompanied by detailed information, including its scientific name, medicinal benefits, and traditional applications in systems such as Ayurveda, Yoga, Unani, Siddha, and Homeopathy. The navigation structure was refined to ensure users could easily explore the plants, with features such as zoom. Usability testing was conducted with initial users, who provided feedback on interface design, responsiveness, and ease of use. Performance optimization techniques were implemented to ensure smooth loading times, especially for VR elements, making the platform accessible across different devices.

Sprint 3: Medicine & Yoga Pose Recommendation Module

The Ayurvedic medicine recommendation module was developed to suggest herbal treatments based on user-input symptoms and health conditions. Data was sourced from classical Ayurvedic texts and modern research to ensure the recommendations were medically relevant and scientifically backed. The system was designed to retrieve pre-stored images and details from the database, ensuring that users received accurate information based on their health conditions.

The Yoga pose recommendation module was developed to provide personalized yoga suggestions based on user health conditions and wellness goals.

Yoga pose recommendations were categorized into three difficulty levels: Beginner, Moderate, and Advanced, allowing users to choose suitable exercises. An early prototype was tested by selected users, and their feedback was incorporated to fine-tune the recommendation logic and UI/UX experience.

Sprint 4: Herbal Teas & QR Code Generation Module

A specialized herbal teas module was introduced, providing insights into various herbal tea recipes, their health benefits, and preparation techniques. The module was designed to categorize herbal teas based on their effects, such as immunity-boosting, digestion-improving, and stress-relieving properties. QR code generation was added, allowing users to easily download plant-related information and medicine recommendations for offline use. User feedback was incorporated to refine the interface, accessibility, and categorization of herbal tea options.

Sprint 5: Advanced Search & Accessibility Features

A voice-based search functionality was developed to enhance accessibility, allowing users to search for information via voice commands instead of typing. Text-to-speech (TTS) was integrated to improve inclusivity, enabling visually impaired users to listen to plant descriptions and recommendations. An advanced search and filter system was implemented, enabling users to sort information based on criteria such as plant category, medicinal properties, symptoms, and traditional usage. Rigorous usability testing was conducted to refine search functionality, ensuring accurate and relevant results. Performance improvements were made to optimize search speed, making information retrieval efficient and seamless for users.

Sprint 6: Multi-Language Support and Final Enhancements

A multi-language translation system was implemented, allowing users to access content in multiple languages to cater to a diverse audience. A user feedback system was developed to collect valuable insights from users, enabling continuous improvement and enhancement of system features.

CHAPTER 2

LITERATURE SURVEY

Ambarwari et al. (2020) proposed “**Plant Species Identification using Leaf Venation Features and SVM.**” This study focuses on identifying plant species based on leaf venation characteristics extracted from binary images obtained through leaf segmentation. The method analyzes features such as skeleton length, number of branches and endpoints, straightness, direction, length ratio, and scale projection, resulting in a total of 19 extracted features. A Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel was used for classification, with 75% of the data used for training. The model achieved an accuracy of 82.67%, a recall of 83%, and an average precision of 84%, demonstrating the effectiveness of venation-based feature extraction for plant identification.[1]

Azadnia et al. (2023) proposed “**An Intelligent Vision-Based System for Medicinal Plant Identification using Convolutional Neural Networks (CNN).**” This study presents a deep learning-based approach for identifying herbal plants through image analysis. The proposed model consists of a convolutional neural network (CNN) block for feature extraction and a classifier block for feature classification. The classifier includes a Global Average Pooling (GAP) layer, a dense layer, a dropout layer, and a SoftMax layer. The system was evaluated using three different image resolutions (64×64, 128×128, and 256×256 pixels) on five distinct medicinal plant species. The model achieved an accuracy of over 99.3%, demonstrating its effectiveness. The study suggests that this approach can serve as a reliable alternative to conventional methods for real-time medicinal plant identification.[2]

Jain, A., et al. (2024) proposed “**Virtual Herbal Garden: An Interactive Platform for Medicinal Plant Exploration in the AYUSH Sector.**” This study presents an interactive system designed to provide detailed information about medicinal plants, including their scientific and local names, cultivation techniques, and medicinal applications. The platform integrates multimedia elements such as high-definition images, videos, and audio content to enhance

user understanding and engagement. Additional features like search functionality and annotations improve accessibility and usability.[3]

Jithendra Kumar and Pradhan, M. R., (2023) developed a **deep learning-based approach for medicinal plant identification using leaf images**. The proposed system utilizes Convolutional Neural Networks (CNNs) to automatically extract features and classify plant species based on leaf characteristics such as shape, texture, and vein patterns. The model was trained and evaluated on a dataset containing various medicinal plant leaves, where preprocessing techniques like resizing and normalization were applied to improve performance. The architecture includes multiple convolutional layers, pooling layers, and fully connected layers for effective feature learning and classification. Experimental results demonstrated high classification accuracy, showing that the deep learning model outperforms traditional machine learning methods. Therefore, the proposed approach provides an efficient and reliable solution for automatic medicinal plant identification.[4]

Kavitha et al. (2023) proposed “**A Real-Time Medicinal Plant Identification System using Mobile Net.**” This study focuses on identifying six medicinal plant species—betel, curry, Tulsi, mint, neem, and Indian beech—using a dataset of 500 images per species collected from Kaggle. The images were preprocessed through resizing and data augmentation techniques to improve model performance. The Mobile Net-based deep learning model achieved an accuracy of 98.33%. The system was further integrated into a mobile application, enabling users to perform real-time plant identification and access information about medicinal benefits. However, the study is limited by its focus on only six plant species and its dependency on high-quality images for accurate predictions.[5]

Mali and Sarode, (2020) developed a **machine learning-based system for identifying medicinal plants**. The proposed model uses image processing techniques along with classification algorithms to recognize plant species based on leaf features such as shape, color, and texture. Feature extraction methods were applied to improve classification accuracy, and multiple machine learning classifiers were tested. The results showed that the system can effectively identify medicinal plants, providing a simple and cost-effective alternative to manual identification methods.[6]

Pawar and Jadhav, (2024) introduced a **mobile-based application for plant identification integrated with Ayurvedic remedies**. The system allows users to capture plant images using a smartphone, after which image processing and machine learning techniques are used to identify the plant species.

Additionally, the application provides detailed Ayurvedic medicinal uses and remedies related to the identified plants. The proposed solution enhances accessibility to traditional knowledge and supports users in real-time plant identification and healthcare guidance.[7]

Patil and Raut, (2022) developed a **smart herbal assistant system that maps diseases to medicinal plants**. The system uses a knowledge-based approach combined with machine learning techniques to recommend suitable medicinal plants for specific diseases. The architecture includes modules for disease input, plant mapping, and recommendation generation. The results demonstrated that the system can effectively assist users in selecting appropriate herbal remedies, making it useful for healthcare and educational applications.[8]

Patil and Kharade, (2021) proposed a **CNN-based model for plant identification and detection of medicinal uses**. The system utilizes deep learning techniques to automatically extract features from plant images and classify them into different categories. The architecture includes convolutional layers, pooling layers, and fully connected layers for accurate prediction. Along with identification, the system also provides information about medicinal uses of the plants. Experimental results indicated improved accuracy compared to traditional approaches.[9]

Prasvita and Herdiyeni, (2020) developed **Med Leaf, a mobile-based medicinal plant identification application using Local Binary Pattern (LBP) and Probabilistic Neural Network (PNN)**. The system extracts texture features using LBP and performs classification using PNN to identify plant species. The application is designed for mobile platforms, enabling users to identify plants easily in real-time. The results showed that the proposed method achieves good accuracy and is efficient for practical usage.[10]

Purandheswari et al., (2023) introduced an **AI-powered AYUSH Virtual Herbal Garden that integrates plant identification and remedy suggestion modules**. The system combines deep learning techniques for plant recognition with a recommendation system for suggesting medicinal uses and remedies. It

provides an interactive platform where users can explore various medicinal plants virtually. The results demonstrated that the system enhances user engagement and promotes awareness of AYUSH-based traditional medicine through intelligent and user-friendly design.[11]

Shelke, K., et al. (2022) proposed “**MedLeaf: A Mobile Application for Medicinal Plant Identification using Leaf Images.**” This study introduces an Android-based application that identifies medicinal plants by analyzing leaf images captured through smartphones. The methodology involves image preprocessing, texture feature extraction using Local Binary Pattern Variance (LBPV), and classification through a Probabilistic Neural Network (PNN). The dataset consists of 1,440 leaf images from 30 Indonesian medicinal plant species. The system achieved an accuracy of 56.33%. However, its performance is limited due to low accuracy caused by variations in image quality and similarities in leaf textures among different species, which affects its reliability and scalability.[12]

2.1 EXISTING SYSTEM

The Existing System of the AYUSH Virtual Herbal Garden is a comprehensive web-based platform designed to educate users about AYUSH medicinal plants. Built with a frontend utilizing HTML, CSS, and JavaScript, and a backend powered by Node.js and MongoDB, the system offers a seamless and interactive learning experience. The architecture is structured to ensure efficient user engagement and data management. It comprises an admin module that facilitates user validation, ensuring secure access and appropriate content management. The user module enables profile management and virtual exploration, allowing individuals to navigate the digital garden effortlessly.

A key component of the system is the Herbal Plants module, which provides extensive details on various medicinal plants, including their scientific and local names, medicinal uses, and propagation methods.

Additionally, the platform features a tab, offering valuable medicinal insights for users interested in alternative healing practices. Registered users can log in to access a rich repository of plant data, accompanied by multimedia content, enhancing their understanding of AYUSH herbal medicine. With an intuitive interface and well-organized content, the platform ensures an engaging and

informative experience for learners, researchers and enthusiasts alike. The user module enables profile management and virtual exploration, allowing individuals to navigate the digital garden effortlessly. A key component of the system is the Herbal Plants module, which provides extensive details on various medicinal plants, including their scientific and local names, medicinal uses, and propagation methods.

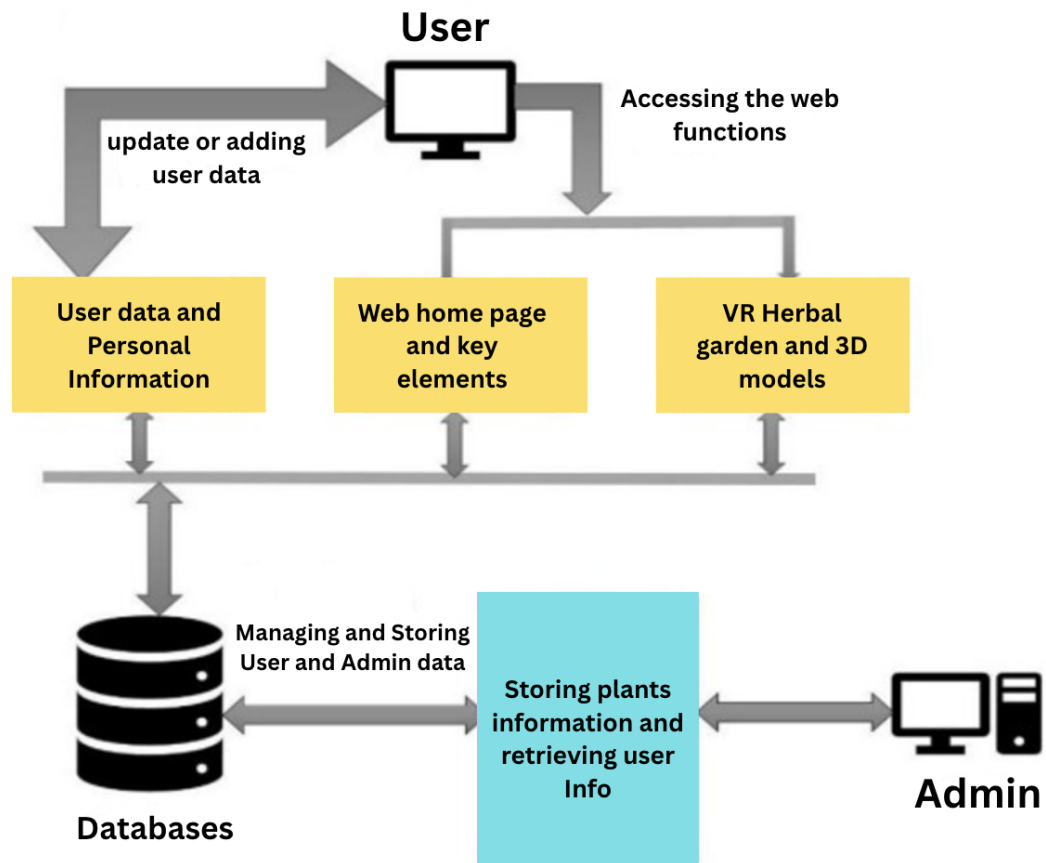


Fig 2.1 Existing System Architecture

The Existing system of real-time identification of medicinal plants involves six plants such as Betel, Curry, Tulsi, Mint, Neem, and Indian Beech—using the MobileNet deep learning model. The existing system’s architecture is designed in three key stages to ensure accurate identification of medicinal plants. First, the MobileNet model is trained over 15 cycles (epochs) to improve its learning. During this stage, 350 images per plant species are used for training, helping the model recognize patterns effectively.

Next, 100 images per species are set aside for validation, which helps fine-tune the model and prevent errors. Finally, the last stage involves testing the model with 50 images per species to check how well it performs on new, unseen data. Through this process, the model is optimized, and after extensive testing, it achieves an impressive 98.33% accuracy, ensuring reliable plant identification for users.

Limitations of Existing Systems:

Current methods for exploring and identifying AYUSH medicinal plants rely heavily on physical gardens or human expertise, which are inconsistent and inaccessible to many. While some digital tools use Convolutional Neural Networks (CNNs) to classify plant species based on leaf images, they lack the immersive and educational depth needed for AYUSH-specific learning. These systems face several limitations:

1. Limited Accessibility:

- Physical gardens and expert-led identification remain inaccessible to remote or non-specialist users.
- The CNN-based system (MobileNet) is limited to six species and requires cloud connectivity, challenging for users without reliable internet.

2. Scalability Issues:

- The CNN tool's dataset of 3,000 images for six species restricts its ability to expand to more plants or adapt to diverse conditions.
- MobileNet's lightweight design limits its capacity for complex tasks, requiring significant resources for scaling.
- The Virtual Herbal Garden's incomplete database and unimplemented features hinder its potential to cover a wide range of plants.

3. Lack of Engagement:

- The CNN system provides basic identification (plant name and benefits) without interactive or multimedia-rich experiences.
- The Virtual Herbal Garden, despite educational goals, lacks 3D/VR immersion in its current prototype, reducing user engagement.

4. Insufficient Context:

- The CNN-based tool focuses solely on identification, omitting comprehensive AYUSH-specific details like medicinal uses or remedies.

- The Virtual Herbal Garden, while aiming to educate, doesn't yet offer detailed Ayurvedic insights or practical applications in its present form.

2.2 GAP ANALYSIS

The existing systems for AYUSH medicinal plant exploration—the web-based Virtual Herbal Garden and the CNN-based real-time identification system using MobileNet—lack the depth and functionality needed for comprehensive AYUSH learning. The Virtual Herbal Garden, built with HTML, CSS, JavaScript, Node JS, and MongoDB, offers basic plant data, limiting engagement. The MobileNet system, with 98.33% accuracy on images of six species, is restricted by its narrow scope, cloud dependency, and lack of AYUSH-specific medicinal context or interactivity. Both systems fail to scale to more species, offer practical features like remedy or yoga recommendations, or include accessibility tools like multi-language support and voice-based search. The Ayush Virtual Herbal Garden fills these gaps with AI-driven identification (EfficientNet, ResNet) for 30 species, and features like medicine recommendations, multi-language translation, and a herbal teas module, ensuring a more engaging and accessible platform.

2.3 RELEVANCE OF THE PROJECT

The Ayush Virtual Herbal Garden holds significant relevance in today's context, where there is a growing global demand for natural remedies and holistic healthcare practices rooted in AYUSH traditions (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy). However, the inaccessibility of physical herbal gardens, the complexity of plant identification, and the lack of comprehensive AYUSH-specific educational tools create barriers for students, practitioners, and enthusiasts.

This project bridges these gaps by leveraging modern technology—By covering 30 plant species and offering features like medicine and yoga pose recommendations, a herbal teas module, multi-language translation, voice-based search, and QR code generation, the system not only simplifies plant identification but also promotes practical application and cultural preservation of AYUSH practices. Its relevance is further underscored by its potential to educate a global audience, empower rural practitioners with digital tools, and support the integration of traditional medicine into mainstream healthcare, aligning with global health trends and the need for inclusive, technology-driven solutions.

CHAPTER 3

AIM AND SCOPE OF PRESENT INVESTIGATION

3.1 REFINED PROBLEM STATEMENT

The AYUSH sector, which includes Ayurveda, Yoga, Unani, Siddha, and Homeopathy, plays a vital role in promoting natural and holistic healing practices. These traditional systems rely extensively on medicinal plants for treatment, prevention, and overall well-being. However, access to physical herbal gardens is often limited due to geographical constraints, lack of infrastructure, and time restrictions. As a result, students, researchers, and enthusiasts face difficulties in exploring and understanding these medicinal plants in a practical manner.

Another major challenge is the accurate identification of medicinal plants. Correct identification requires specialized botanical knowledge, which is not commonly available to the general public or even to many practitioners. This knowledge gap often leads to misidentification of plants, resulting in incorrect usage, reduced effectiveness of remedies, and in some cases, potential health risks. Consequently, this affects the reliability and trust in traditional AYUSH-based treatments.

Existing solutions such as textbooks, research articles, and online databases primarily provide theoretical information. While informative, they lack interactivity, personalization, and practical guidance. Many currently available plant identification systems are designed for general botanical purposes and do not focus specifically on AYUSH medicinal plants. Furthermore, these systems often have limited datasets, restricted plant coverage, and do not provide additional contextual features such as medicinal recommendations, usage guidelines, or integration with traditional practices like yoga.

To address these limitations, the proposed Ayush Virtual Herbal Garden system introduces an innovative, AI-powered digital platform that enhances accessibility, usability, and learning experience. The system integrates advanced deep learning models such as EfficientNet and ResNet to accurately identify up to 30 medicinal plant species from user-uploaded images. This significantly improves plant recognition compared to traditional systems with limited scope.

In addition to plant identification, the platform provides detailed information about each plant, including scientific names, medicinal benefits, and traditional applications across AYUSH systems.

To further enhance user engagement and practical utility, the system includes features such as medicine recommendations based on identified plants, a herbal teas module, and integration of AYUSH knowledge for better understanding of natural remedies.

The platform is designed with accessibility in mind, offering multi-language translation support and user-friendly navigation, making it suitable for a wide range of users including students, researchers, and common users. By combining artificial intelligence with traditional knowledge, this system bridges the gap between ancient herbal wisdom and modern technology.

Overall, the Ayush Virtual Herbal Garden not only simplifies medicinal plant identification but also provides an interactive, informative, and accessible environment for learning and applying AYUSH principles, thereby contributing to the preservation and promotion of traditional healthcare systems.

3.2 SCOPE

The Virtual Herbal Garden will serve as a comprehensive digital platform designed to educate and engage users with medicinal plants and their applications. Users will be able to interact with plants, and access detailed information about their scientific classification, benefits, and traditional uses. The platform will include an AI-driven plant identification feature, allowing users to upload plant images and obtain reliable insights into their medicinal properties.

Beyond identification, the system will act as a holistic wellness guide, offering personalized recommendations for herbal remedies and yoga postures to address various health concerns. To improve usability, the platform will support search and filter options, voice-based search, text-to-speech features, and multi-language translation, ensuring accessibility for a diverse audience. Additionally, users can generate and download QR codes for plant profiles, facilitating easy sharing and quick reference. The platform will also include a Herbal Teas Module, educating users on the benefits and preparation of traditional herbal drinks for better health.

With its blend of interactive experiences, AI-powered identification, and holistic wellness integration, the Virtual Herbal Garden will be a valuable educational resource for students, practitioners, and health enthusiasts. In the future, it can be expanded with augmented reality (AR) features, expert consultations, and an extended herbal database, further enhancing its impact on natural medicine and healthcare awareness.

3.3 PROPOSED SYSTEM

Ayush Virtual Herbal Garden is an advanced solution that seamlessly integrates AI-driven plant identification with an interactive virtual platform, specifically designed to overcome the limitations of existing systems for AYUSH (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy) medicinal plants. Leveraging Transfer Learning, the system utilizes pre-trained deep learning models such as EfficientNet and ResNet to achieve superior identification accuracy and computational efficiency, trained on a diverse dataset of 2,000 leaf images representing 30 AYUSH plant species. The Flask-based web application serves as the backbone of the system, providing a user-friendly interface that delivers comprehensive information on medicinal plants, including their botanical names, natural habitats, Ayurvedic medicinal properties, and cultivation techniques. This immersive space allows users to navigate a digital herbal garden and interact with high-quality plant images accompanied by detailed descriptions. Additionally, the system offers features such as disease remedy suggestions based on Ayurvedic principles, multimedia content including audio guides, and a search functionality that enables users to filter plants by medicinal use or species type. This holistic approach ensures that the platform serves as a robust, educational, and practical tool for exploring AYUSH herbal knowledge.

The application is structured into five key modules, each designed to enhance accessibility and provide valuable insights. The Tour module offers a virtual garden tour, allowing users to experience an interactive herbal environment, simulating a real-world exploration. The Herbal Info module provides detailed information on 30 medicinal plant species, sourced from the dataset, including their traditional uses and health benefits.

Advantages of Proposed System:

- 1.Enhanced Accuracy:** AI ensures precise identification, even for visually similar AYUSH plants.
- 2.Efficient Plant Identification:** The system allows users to quickly identify medicinal plants by uploading leaf images, reducing manual effort and time.
- 3.Comprehensive Knowledge:** Detailed AYUSH-specific insights, including medicinal uses and remedies are readily available
- 4.Accessibility and Scalability:** The web-based platform and efficient Transfer Learning make it usable by a wide audience with minimal resources

3.4 System Architecture

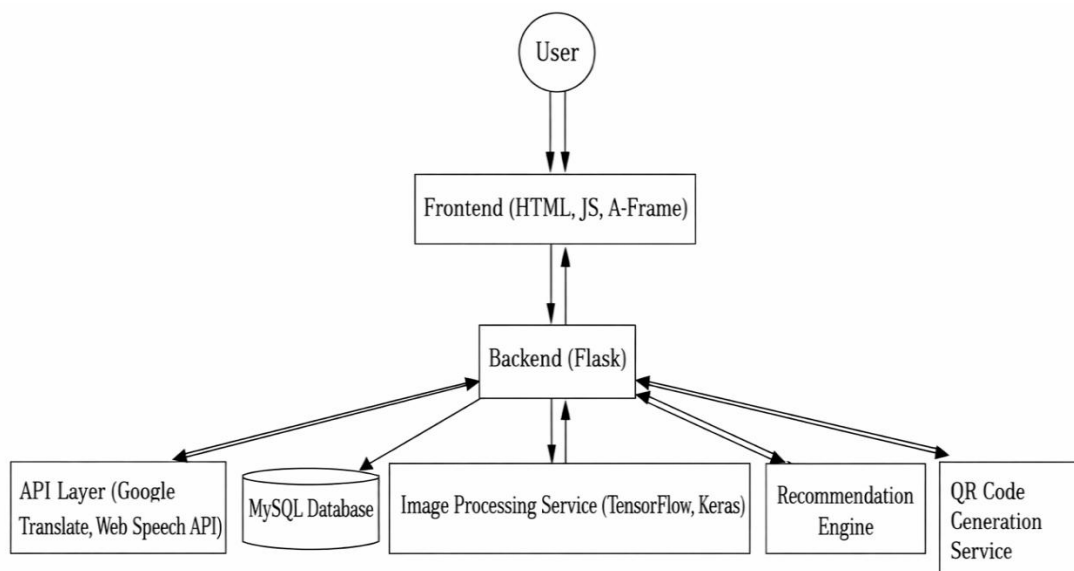


Fig: 3.1 Proposed System Architecture Diagram

The above fig 3.1 represents proposed system architecture. It is designed to provide an interactive and immersive experience for users exploring medicinal plants and Ayurvedic remedies. The frontend sends HTTP requests to the backend, developed using Flask, which handles data retrieval, plant identification, medicine recommendations, and yoga posture suggestions. The backend communicates with a MySQL database using SQLAlchemy to fetch and store plant data, user feedback, and herbal tea recipes. Users can upload plant images for identification, where the image is processed by a TensorFlow and Keras model to recognize the plant species. Additionally, the system provides personalized Ayurvedic recommendations using a plant-remedy mapping stored in the database. Features like voice-based search, text-to-

speech, and multi-language translation are implemented using the Web Speech API and Google Translate API.

Users can also generate and download QR codes for plant information using a QR Code Generation Service. The seamless interaction between the frontend, backend, database, and external APIs ensures an engaging user experience while promoting knowledge about traditional herbal medicine. The application is structured into five key modules, each designed to enhance accessibility and provide valuable insights. The Tour module offers a virtual garden tour, allowing users to experience an interactive herbal environment, simulating a real-world exploration.

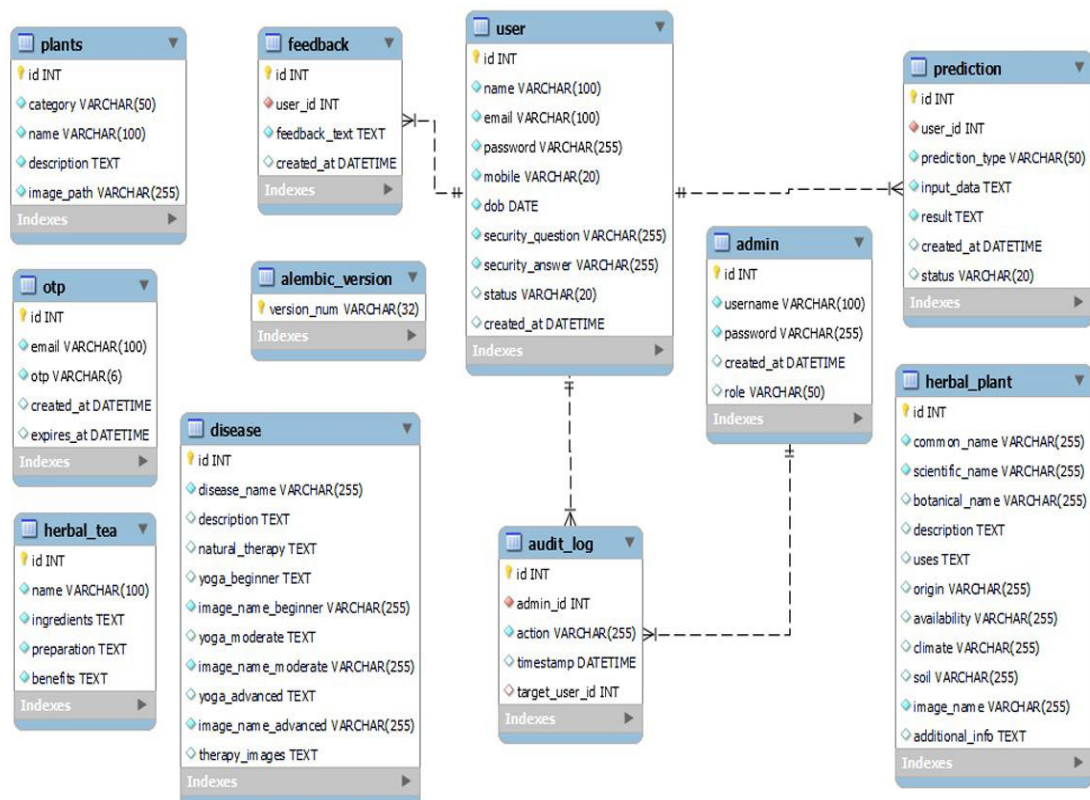


Fig: 3.2 ER Diagram

1. The User entity stores user details and has a one-to-many relationship with Prediction and Feedback.
2. The Admin entity manages the system and has a one-to-many relationship with Audit_Log.

3. The Audit_Log entity tracks admin actions and is linked to Admin.
4. The OTP entity is linked to User via email for authentication purposes.
5. The Feedback entity stores user feedback and has a many-to-one relationship with User.
6. The Prediction entity records plant-related predictions and has a many-to-one relationship with Users.
7. The Alembic Version entity tracks database schema versions and is independent.
8. The Plants entity stores plant details and has no direct relationships.
9. The Herbal Plant entity contains information on medicinal plants and is not linked to other tables.
10. The Herbal Tea entity records herbal tea details and has no direct relationships.
11. The Disease entity stores disease-related information and is not directly connected to any other tables.

The frontend sends HTTP requests to the backend, developed using Flask, which handles data retrieval, plant identification, medicine recommendations, and yoga posture suggestions. The backend communicates with a MySQL database using SQLAlchemy to fetch and store plant data, user feedback, and herbal tea recipes. Users can upload plant images for identification, where the image is processed by a TensorFlow and Keras model to recognize the plant species. Additionally, the system provides personalized Ayurvedic recommendations using a plant-remedy mapping stored in the database. Features like voice-based search, text-to-speech, and multi-language translation are implemented using the Web Speech API and Google Translate API.

Users can also generate and download QR codes for plant information using a QR Code Generation Service. The seamless interaction between the frontend, backend, database, and external APIs ensures an engaging user experience while promoting knowledge about traditional herbal medicine.

The system must be able to classify Ayurvedic medicinal plants using Convolutional Neural Networks (CNN). It should leverage a dataset of around 2,000 images from 30 species of medicinal leaves for accurate classification. The module should use Transfer Learning to enhance accuracy and efficiency

in plant identification. Features like voice-based search, text-to-speech, and multi-language translation are implemented using the Web Speech API and Google Translate API.

Relationships Between Tables

User & Prediction → One-to-Many

1. user_id in Prediction references id in User.
2. User (1) → (∞) Prediction

User & Feedback → One-to-Many

1. user_id in Feedback references id in User.
2. User (1) → (∞) Feedback

Admin & Audit Log → One-to-Many

1. arminid in Audit Log references id in Admin.
2. Admin (1) → (∞) Audit Log

User & OTP → One-to-One (via email)

1. email in OTP indirectly links to email in User.
2. User (1) → (1) OTP

Roles & Activities:

1. User:

- Registers an account.
- Generates and validates OTP for authentication.
- Logs into the system.
- Submit feedback.
- Uses the system to make predictions (likely related to plant identification or herbal recommendations).

2. Admin:

- Logs into the system.
- Manages users (possibly approving registrations or handling user queries).
- Manages content (updating plant/herbal data, adding information, or moderating discussions).
- Views audit logs (tracking activities for security and analytics).

3. Disease Module:

- Retrieves disease details (fetching information about ailments).
- b. Provides herbal therapy recommendations based on the disease.

- c. Suggests yoga practices categorized by difficulty level.

4. Herbal Plant Module:

- Retrieves plant details (scientific name, medicinal uses, etc.).
- b. Allows users to search for specific plants.

5. Herbal Tea Module:

- Retrieves details about herbal teas (ingredients, benefits, etc.).
- b. Allows users to search for teas based on preferences or health benefits.

The system integrates user, admin, and multiple functional modules to provide a unified platform for medicinal plant identification and herbal healthcare support. Users can securely access the system, make predictions, and give feedback, while admins manage data and monitor activities. The disease, herbal plant, and herbal tea modules provide relevant information, recommendations, and search features, making the system efficient and user-friendly.

This system is designed to automate the process of medicinal plant identification using advanced deep learning techniques. It utilizes image classification models to analyze leaf images and accurately determine the plant species. The workflow involves data preparation, model training, evaluation, and deployment through a user interface, ensuring smooth interaction for end users. By integrating machine learning with a structured pipeline, the system enhances accuracy, efficiency, and scalability in plant recognition tasks, making it suitable for applications in healthcare, education, and environmental studies. The system emphasizes the importance of data-driven decision-making by leveraging machine learning techniques for accurate prediction and analysis. It follows a structured approach that includes data acquisition, preprocessing, model training, and performance evaluation to ensure reliable outcomes. The integration of deep learning models enhances the capability of the system to handle complex visual patterns in leaf images. Furthermore, the deployment of the model through an interactive platform improves usability, allowing users to easily access predictions and related information. This approach not only increases efficiency but also supports scalable and real-time implementation in various domains.

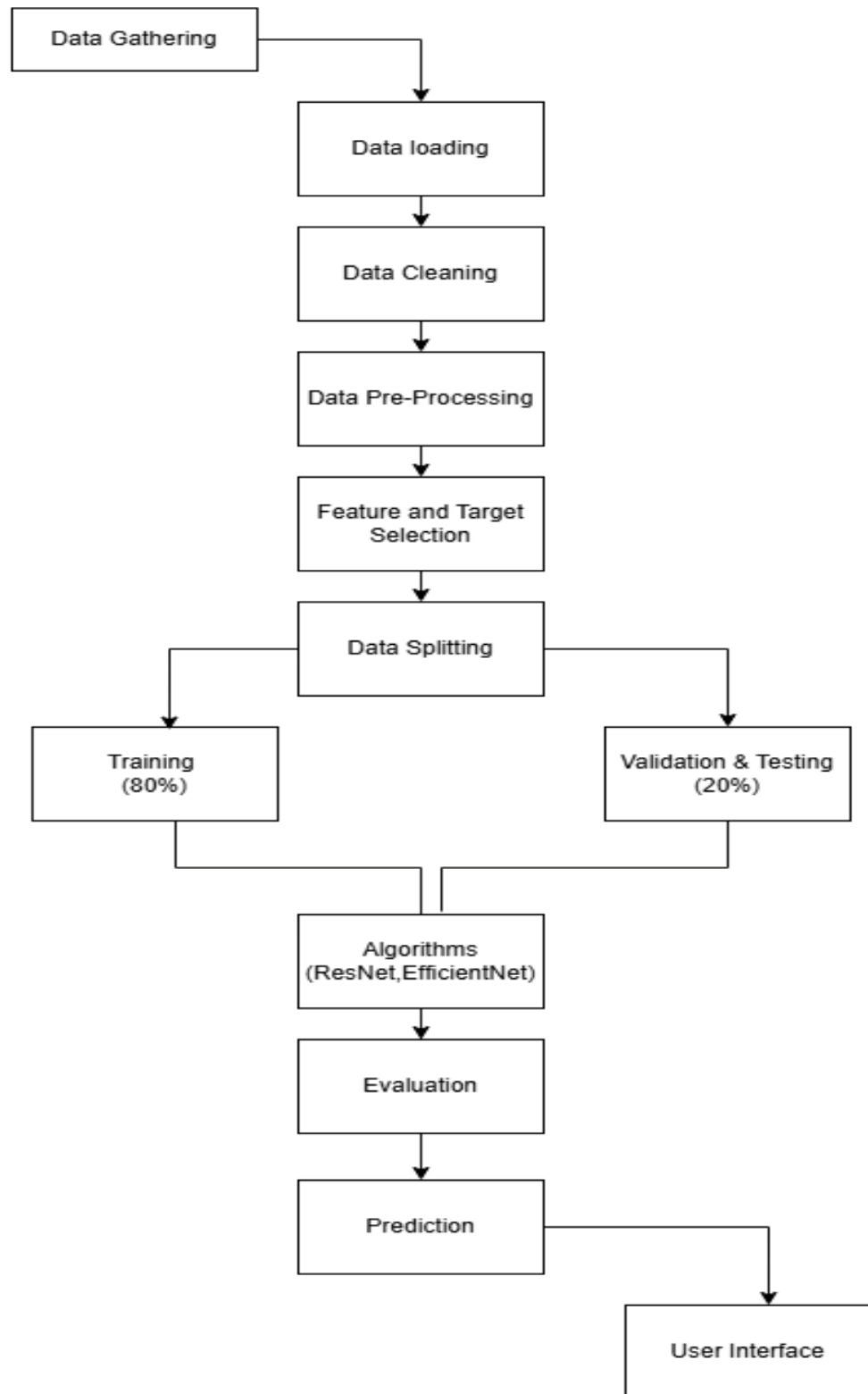


Fig: 3.3 Block Diagram

Fig: 3.3 System Workflow for Herbal Plant Prediction represents the complete architecture and step-by-step pipeline of the Ayush Virtual Herbal Garden system, integrating data processing, deep learning, and user interaction. The workflow ensures that raw data is systematically transformed into meaningful predictions that users can easily access through the interface. The process begins with data gathering, where a diverse collection of herbal leaf images is collected from datasets and real-world sources. This ensures that the model learns from varied patterns and improves accuracy. The data loading phase imports this data into the system environment for processing. During data cleaning, unwanted noise, duplicate entries, and corrupted images are removed to maintain high-quality input data. In the data preprocessing stage, images are resized, normalized, and sometimes augmented to make the model more robust. The feature and target selection step identifies important visual characteristics of leaves and assigns correct labels. After this, data splitting divides the dataset into training (80%) and validation/testing (20%) sets to avoid overfitting and ensure proper evaluation. The training phase uses deep learning models like ResNet and EfficientNet, which are highly effective for image classification tasks. These models learn complex patterns from the training data. The validation and testing phase checks how well the model performs on unseen data. In the evaluation stage, metrics such as accuracy, precision, and loss are used to measure performance and reliability. Finally, in the prediction phase, the trained model is used to classify new leaf images uploaded by users. The results are then displayed through the user interface, providing an interactive and user-friendly experience.

Key Points:

- Ensures systematic flow from raw data to final prediction
- Uses data cleaning and preprocessing for improved accuracy
- Implements data splitting to avoid overfitting
- Utilizes advanced models like ResNet and EfficientNet
- Provides evaluation metrics to measure model performance
- Supports real-time prediction for user-uploaded images
- Integrates with a user-friendly interface for easy interaction

Overall, this workflow demonstrates a well-structured and efficient approach to building a robust herbal plant prediction system using machine learning and computer vision techniques.

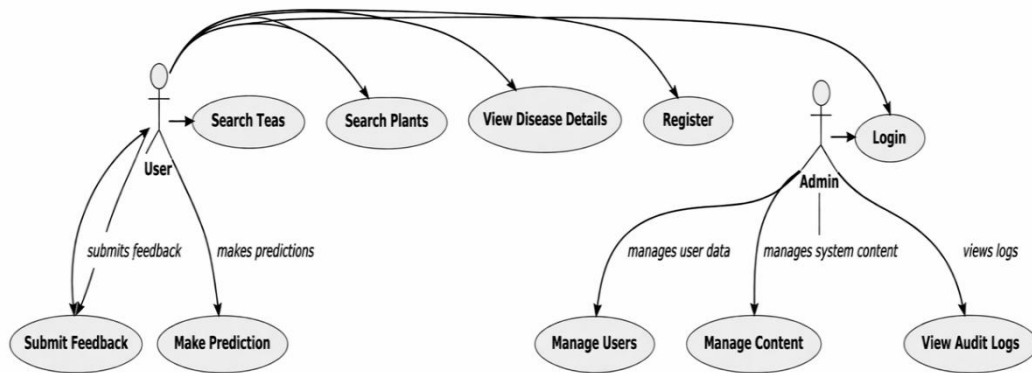


Fig: 3.4 Use Case Diagram

Actors:

1. User: A general user who can register, login, explore herbal plants, search for diseases, check herbal tea information, and submit feedback
2. Admin: Manages users, handles content, views audit logs, and monitors activities.

Use Cases for Users:

1. Register: Users register by providing their details like name, email, password, and mobile number.
2. Login: Users log in using their email and password.
3. Explore Herbal Plants: Users can search for detailed information about herbal plants.
4. Explore Diseases: Users can view disease-related information and suggested therapies.
5. View Herbal Teas: Users explore different herbal teas, their benefits, and preparation.
6. Submit Feedback: Users can submit feedback or suggestions about the system.

Use Cases for Admins:

1. Login: Admins login using admin credentials.
2. Manage Users: Admins can view, edit, or remove users from the system.

3. Manage Content: Admins can add, update, or delete plant, tea, or disease information.
4. View Audit Logs: Admins can view records of actions performed in the system for monitoring purposes.

CHAPTER 4

EXPERIMENTAL OR MATERIALS AND METHODS; ALGORITHMS USED

4.1 FUNCTIONAL REQUIREMENTS:

Plant Identification Module

The system must be able to classify Ayurvedic medicinal plants using Convolutional Neural Networks (CNN). It should leverage a dataset of around 2,000 images from 30 species of medicinal leaves for accurate classification. The module should use Transfer Learning to enhance accuracy and efficiency in plant identification.

User Interface (UI) via Flask

The system must provide a Flask-based web interface for users to interact with the plant identification module. The UI should display detailed plant information, including scientific name, medicinal uses and geographical details.

Disease and Remedy Module

The system should include a module that allows users to input symptoms or diseases. Based on the input, it should suggest Ayurvedic remedies using a predefined knowledge base of medicinal plants.

Prediction Module

The module must utilize AI-powered predictions to identify plant species based on uploaded images. It should provide scientific and geographical details about the identified plant.

Herbal Teas Module

The system must include a module dedicated to herbal teas, offering recipes and benefits based on Ayurvedic medicinal plants. The module must display preparation ingredients, instructions and health benefits for each herbal tea recipe.

Accessibility and User Engagement

The system should ensure easy navigation and accessibility for users with varying levels of expertise. It must provide interactive elements such as search functionalities, plant comparison, or bookmarking of plants for future reference.

Reliability and Performance

The AI model should achieve high classification accuracy, minimizing false predictions.

The system should efficiently handle multiple requests and process plant identifications in real time.

Data Management

The system must store and manage plant images, descriptions, and remedy information effectively. It should allow for future updates and scalability in adding new plant species and Ayurvedic remedies.

4.2 SOFTWARE AND HARDWARE REQUIREMENTS:

Hardware:

- Personal computer or laptop with sufficient RAM (8 GB or higher recommended)
- Adequate storage space for datasets and model files
- Processor capable of handling computational tasks involved in model training

Software:

- Python (programming language), Anaconda
- Jupyter Notebook (for data analysis and model development)
- Scikit-learn (machine learning library)
- Pandas (data manipulation library)
- NumPy (numerical computing library)
- Matplotlib and Seaborn (data visualization libraries)

4.3 FEASIBILITY STUDY:

1. Technical Feasibility

The project uses Flask for the web application and EfficientNet and ResNet for AI-based plant identification. A-Frame is integrated for creating an interactive virtual herbal garden. The AI models are chosen for their accuracy and compatibility with the system. Cloud storage ensures efficient data management for images and plant information. Features like voice search, text-to-speech, and multilingual support are technically achievable.

2. Economic Feasibility

The project uses open-source tools like Flask and Python, reducing development costs. Pre-trained AI models minimize the expenses of training from scratch. Cloud hosting was selected for its cost-effectiveness and scalability. The project provides social benefits by promoting awareness of AYUSH medicinal plants.

3. Operational Feasibility

Users like students, researchers, and AYUSH practitioners can easily access the system. The plant identification system provides real-time results with high accuracy. The virtual tour enhances user engagement through immersive experiences. Voice search and multi-language support ensure accessibility for diverse users. A feedback system allows users to provide suggestions, improving operational effectiveness. Admins can manage the plant database, ensuring up-to-date information.

4.4 ALGORITHMS USED

ResNet (Residual Network) introduced residual learning with skip connections, solving the vanishing gradient problem that occurs in very deep networks. It is extensively used in medical image diagnosis, face recognition, and self-driving car vision systems. ResNet assumes that deeper networks should ideally perform better, but without residual connections, training deep networks becomes difficult. Its key advantage is its ability to train very deep networks while maintaining performance, though it requires significant computational resources.

Step 1: Data Collection and Preprocessing

Collected a dataset of 2,000 leaf images from 30 plant species. Images resized to 224x224 pixels using Pillow. Images normalized by scaling pixel values to the range [0,1]. Data augmentation applied using random flipping, rotation, zooming, and rescaling for improved generalization.

Step 2: Model Selection and Initialization

Selected ResNet50V2 as the base model for transfer learning. Loaded pre-trained weights from ImageNet. Initial layers were frozen to retain learned features.

Step 3: Model Customization

Applied global average pooling to reduce the spatial dimensions and minimize parameters. Added a dense layer with 30 neurons (equal to the number of plant species) with softmax activation for multi-class classification. The model summary showed a total of 23,564,800 parameters.

Step 4: Compilation

Used Sparse Categorical Cross entropy as the loss function for multi-class classification. Implemented Adam optimizer with a learning rate of 0.001. Selected accuracy, precision, recall, and F1-score as evaluation metrics.

Step 5: Learning Rate Adjustment

Applied the Reduce LROn Plateau callback to monitor validation loss and reduce the learning rate when stagnation occurred, preventing overfitting. Data augmentation applied using random flipping, rotation, zooming, and rescaling for improved generalization.

Step 6: Model Training

Split the dataset into 80% training and 20% validation. Trained the model for 20 epochs using a batch size of 32. Implemented early stopping to terminate training if validation loss stopped improving. The model summary showed a total of 23,564,800 parameters.

Step 7: Evaluation

Evaluated model performance using the test dataset.

Table 4.1: ResNet Classification Table

Accuracy	Precision	Recall	F1 score
96%	97%	97%	97%

EfficientNet is a family of convolutional neural networks (CNNs) designed for high-performance image classification tasks with optimized accuracy and computational efficiency. EfficientNet-B0 is the base version, and it uses a compound scaling method to balance depth, width, and resolution for optimal results.

The EfficientNetB0 model was implemented for Ayurvedic medicinal plant classification:

Step 1: Data Collection and Preprocessing Collected a dataset of 2,000 leaf images from 30 plant species. Images resized to 224x224 pixels using Pillow.

Pixel values normalized to the range [0,1] for efficient processing. Applied data augmentation techniques such as random flipping, rotation, and zooming to improve generalization.

Step 2: Model Selection and Initialization Choose EfficientNetB0 as the base model for transfer learning. Loaded pre-trained weights from the ImageNet dataset. Set `include_top=False` to remove the default classification layer.

Step 3: Model Customization Added a `GlobalAveragePooling2D` layer to reduce feature map dimensions using global averaging, making it computationally efficient. Included a dense layer with 128 neurons and ReLU activation (`Dense (128, activation='relu')`) for introducing non-linearity and enhancing learning. Added a final dense layer with 30 neurons and softmax activation for multi-class classification. The model consisted of 4,217,409 total parameters, with 167,838 trainable and 4,049,571 non-trainable parameters.

Step 4: Compilation Compiled the model using the Adam optimizer with a learning rate of 0.001. Choose `sparse_categorical_crossentropy` as the loss function, as it is suitable for integer-encoded multi-class classification. Set accuracy as the primary evaluation metric.

Step 5: Model Training Split the dataset into 80% training and 20% validation. Trained the model using a batch size of 32 for 5 epochs. Applied early stopping to prevent overfitting and save the best-performing model.

Step 6: Evaluation Evaluated the model using the test dataset.

Table 4.2: EfficientNet Classification Table

Accuracy	Precision	Recall	F1 score
92.3	92.3	92.3	92.3

Flask

1. Description: Flask is a lightweight Python web framework that is used for building web applications. It provides essential features such as routing, request handling, and templating without the overhead of larger frameworks.

2. Usage: Manages the core application logic, handles different routes for plant information, herbal teas, plant identification, and user feedback. It integrates with MySQL via SQLAlchemy and uses Jinja2 for rendering dynamic web pages.

SQLAlchemy

1. Description: SQLAlchemy is an ORM (Object-Relational Mapping) tool for Python. It allows interaction with the database through Python objects, eliminating the need for raw SQL queries.
2. Usage: Manages interactions between Flask and MySQL, simplifying database operations like CRUD (Create, Read, Update, Delete) for plant data, herbal teas, and user feedback.

Jinja2

1. Description: Jinja2 is a templating engine for Python, used by Flask to dynamically generate HTML pages.
2. Usage: Renders dynamic plant details, health recommendations, yoga postures, and other content on web pages by combining static HTML with dynamic data.

HTML, CSS, and JavaScript

1. Description: HTML (HyperText Markup Language) provides the structure, CSS (Cascading Style Sheets) defines the design, and JavaScript enables dynamic behavior in web pages.
2. Usage: HTML is used for structuring web pages, CSS for styling the user interface, and JavaScript for interactive features like dynamic plant searches, filtering, and navigating the virtual herbal garden.

Web Speech API

1. Description: The Web Speech API is a browser-based API for speech recognition and speech synthesis.
2. Usage: Implements Voice-Based Search for plant information using speech recognition. It also integrates Text-to-Speech (TTS) to allow users to hear plant details, making the system more accessible.

TensorFlow & Keras

1. Description: TensorFlow is an open-source machine learning framework developed by Google, and Keras is a high-level neural network API. These tools are used for deep learning tasks.
2. Usage: Used for Medicinal Plant Identification through image uploads. The system uses machine learning models (like ResNet50 and EfficientNetB0) to classify plants based on leaf images.

Scikit-learn

1. Description: Scikit-learn is a Python library for machine learning, used for data mining and data analysis.
2. Usage: Handles tasks like splitting datasets into training and testing sets, as well as evaluating machine learning models using metrics like accuracy, precision, and recall.

QR Code Library (qrcode)

1. Description: The qrcode library allows generation of QR codes in Python.
2. Usage: Enables the generation of QR codes for plant information pages, which users can scan or share for quick access to detailed plant data.

MySQL

1. Description: MySQL is an open-source relational database management system.
2. Usage: Stores data such as plant information, medicinal remedies, yoga poses, herbal tea recipes, user feedback, and more. It is connected to Flask using SQLAlchemy for data retrieval and storage.

Flask-SQLAlchemy

1. Description: Flask-SQLAlchemy is an extension for Flask that simplifies database interaction using SQLAlchemy.
2. Usage: Provides seamless integration between Flask and MySQL, facilitating easy database operations related to plant data, remedies, user feedback, etc.

4.5 IMPLEMENTATION

Features

1. Plant Information Module

This module offers comprehensive details about medicinal plants, including their names, scientific classifications, properties, benefits, and traditional uses. Users can visually explore the plants and gain insights into their significance in the AYUSH system, enriching their knowledge of natural Remedies, Flask retrieves plant data from MySQL using SQLAlchemy. Jinja2 templates display plant details with images on the web interface.

2. Medicine Recommendation for Health Issues

Based on the user's input regarding their health concerns, the system recommends suitable herbal medicines and formulations derived from

Ayurvedic classical texts. This feature promotes natural healing by providing personalized solutions aligned with traditional medicine practices. Plant remedy mappings stored in MySQL. Flask queries remedies based on identified plants using SQLAlchemy. Jinja2 templates display remedies.

3. Yoga Posture Suggestions for Health Issue Cure

To promote holistic well-being, the platform suggests appropriate yoga postures that can help alleviate specific health conditions. Yoga poses are stored in MySQL. Flask suggests poses based on disease of the user, Jinja2 template displays poses with images.

4. Medicinal Plant Identification by Image Upload

Users can upload images of plants to identify them using the system's machine learning-based image recognition feature. The platform accurately identifies the plant species and displays relevant information on its medicinal properties and traditional uses, making plant identification accessible to all users. 2,000 leaf images for 30 species are used. Pillow resizes images to 224x224, normalizes, and augments (rotation, flipping). TensorFlow and Keras fine-tune ResNet50 and EfficientNetB0 for 20 and 5 epochs. Scikit-learn splits data into training and validation data set. Matplotlib/Seaborn visualize metrics (accuracy, precision). Flask backend processes user-uploaded images for identification.

5. User Feedback System

The feedback system enables users to share their experiences, suggest improvements, or report issues. This fosters community engagement and helps the development team enhance the platform continuously. User opinions and suggestions are valued to ensure a user-centric experience. HTML form captures user feedback. Flask stores feedback in MySQL using SQLAlchemy.

6. Search and Filter Functionality

Users can efficiently find medicinal plants by using the search and filter options. They can filter plants based on criteria such as name, health benefits, or Ayurvedic classification. This feature ensures quick access to relevant information, improving the overall user experience. Flask queries MySQL for plants by name. Flask enables filtering and updates results.

7. Voice-Based Search Function

The voice-based search feature allows users to search for plant information through voice commands. This makes the platform accessible to users with

disabilities or those who prefer hands-free interaction. It enhances convenience and offers a more inclusive experience. The Web Speech API allows users to search for plants by speaking their queries instead of typing.

The API captures the user's voice input through the browser's microphone and converts it into text using speech-to-text technology. This text is then sent to the Flask backend, which queries the MySQL database to find matching plant information. Finally, the relevant plant details are displayed on the web application interface, providing a faster and more interactive search experience.

8. Text-to-Speech (TTS) Feature

Users can listen to plant descriptions, medicinal uses, and other information using the text-to-speech feature. This is particularly helpful for those with visual impairments or users who prefer auditory learning. The feature enhances accessibility and user engagement. Implemented using Web Speech API, it converts plant info to speech. By clicking on the speaker button on the interface, the text will be read aloud.

9. Multi-Language Translation

To cater to a diverse audience, the platform supports multiple languages, including English, Telugu, and Hindi. Users can switch between languages to read plant information in their preferred language, ensuring a more inclusive and accessible experience. It is implemented using the Google Translate API.

10. QR Code Generation with Download Option

Each medicinal plant profile is equipped with a QR code that users can generate and download. This allows users to share plant information easily with others or quickly access it later using a mobile device, promoting the spread of knowledge about traditional medicine. It is implemented by using the qrcode library in Flask to generate QR codes that link directly to specific plant information pages.

11. Herbal Teas Module

This module educates users about various herbal teas, their preparation methods, and their health benefits. Users can explore different herbal tea recipes made from natural ingredients such as ginger, Tulsi, and mint. The module promotes the use of herbal beverages for health and wellness, inspired by traditional practices. Flask handles the retrieval of relevant recipes from the database, while the front-end interface displays the recipes in an organized and visually appealing manner.

CHAPTER 5

RESULT AND DISCUSSION, PERFORMANCE ANALYSIS

5.1 TESTCASES AND REPORTS

The following test cases were designed and executed to validate the functionality of the project:

1. Importing Libraries:

- a. Objective: Ensure necessary libraries such as TensorFlow, Keras, NumPy, and Seaborn are imported without errors.
- b. Expected Result: Libraries should be successfully imported.
- c. Actual Result: Libraries imported without any issues.
- d. Status: Pass

2. Data Collection and Loading:

- a. Objective: Verify the dataset is accessible and correctly loaded from the specified directory.
- b. Expected Result: Dataset should load successfully.
- c. Actual Result: Data loaded and displayed for visualization.
- d. Status: Pass

3. Data Analysis:

- a. Objective: Perform exploratory data analysis (EDA) to check for class imbalances and visualize data patterns.
- b. Expected Result: Data should be properly analyzed using visualizations.
- c. Actual Result: Visualizations were successfully generated without errors.
- d. Status: Pass

4. Image Pre-processing:

- a. Objective: Verify the resizing of images to 224x224, normalization using rescaling, and augmentation using flipping, rotation, and zooming.
- b. Expected Result: Images should be pre-processed without errors.
- c. Status: Pass
- d. Actual Result: Image pre-processing was successfully completed.

5. Feature Selection and Extraction:

- a. Objective: Ensure essential features are extracted using pre-trained models like ResNet50V2 and EfficientNet.
- b. Expected Result: Features should be successfully extracted from images.

c. Actual Result: Feature extraction completed without errors.

d. Status: Pass

6. Train-Test Split:

a. Objective: Validate the correct splitting of the dataset into 70% training and 30% validation data.

b. Expected Result: Data should be correctly split with balanced class distribution.

c. Actual Result: Data was successfully split into training and validation sets.

d. Status: Pass

7. Model Application:

a. Objective: Evaluate the performance of ResNet50V2 and EfficientNet models.

b. Expected Result: Models should be applied successfully without errors.

c. Actual Result: Both models were evaluated, and results were obtained as expected.

d. Status: Pass

8. Prediction:

a. Objective: Ensure the model correctly predicts the class of unseen images.

b. Expected Result: Model should predict accurate results.

c. Actual Result: Predictions were successfully made and verified.

d. Status: Pass

5.2 OUTPUTS



Fig: 5.1 Home Page

The above Fig. 5.1 Home Page functions as the initial interface of the application, offering users a welcoming and easy-to-use environment. It highlights the platform's purpose through a brief description, helping users understand its features quickly. The design is simple and well-organized to ensure smooth navigation. Essential options like "Login" and "Sign Up" are prominently placed for both existing and new users. These options allow users to access their accounts or create a new account without confusion. Visual elements such as banners or icons enhance the overall look and feel of the page. The layout is designed to guide users efficiently to the next step. Overall, the homepage delivers a smooth and intuitive user experience.

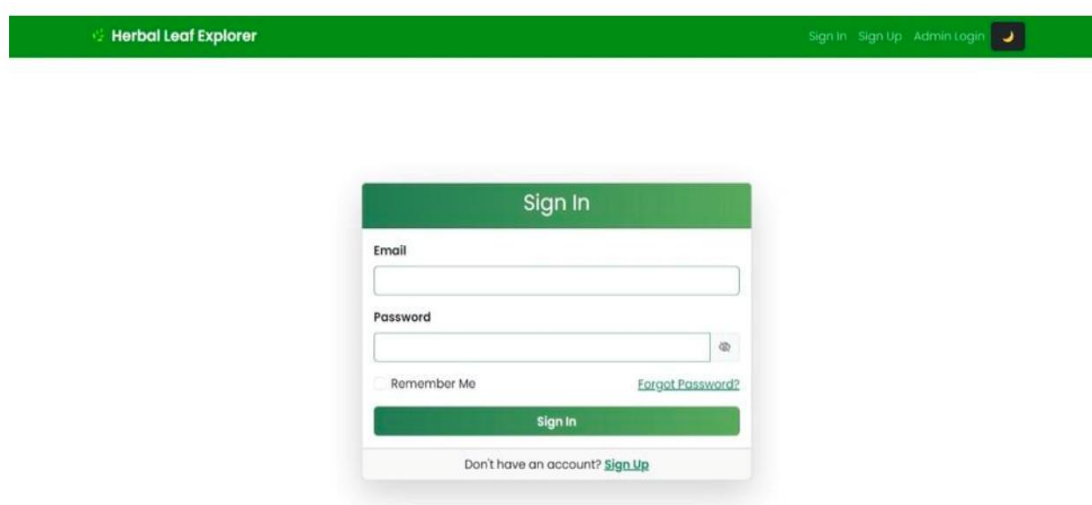


Fig: 5.2 Sign in Page

The above Fig. 5.2 Home Page serves as the primary access point of the application, offering users a simple and engaging interface. It presents the platform's objective through a short introduction or tagline, allowing users to quickly grasp its purpose. The design is maintained in a clean and user-friendly manner to support easy navigation. Important options such as "Login" and "Sign Up" are clearly visible for both existing and new users. These options enable users to securely log in or register without any difficulty. Visual components like icons or banners may be added to improve the interface appearance. The layout is structured to help users move smoothly to the next stage. Overall, the homepage provides a convenient and user-friendly experience.

The screenshot shows the 'Sign Up' page of the 'Herbal Leaf Explorer' application. The page has a green header with the application name and navigation links for 'Sign In', 'Sign Up', and 'Admin Login'. The main form is titled 'Sign Up' and contains several input fields: 'Full Name', 'Email', 'Password' (with a visibility toggle), 'Mobile', 'Date of Birth' (with a date picker), 'Security Question', and 'Security Answer'. The form is styled with a clean, modern design and a soft shadow.

Fig: 5.3 Sign Up Page

The SignUp page in fig 5.3 enables new users to register and access the platform's features. It includes input fields for details like name, email, password, and confirm password, with proper validation for accuracy and security. The interface is simple and user-friendly, ensuring a smooth registration process. Features like password visibility toggle and error messages enhance usability. The frontend is developed using HTML, CSS, and JavaScript or React for better interactivity. The backend uses technologies like Node.js or Python with databases such as MySQL or MongoDB to securely store user information.

The screenshot shows the 'Admin Login' page of the 'Herbal Leaf Explorer' application. The page has a green header with the application name and navigation links for 'Sign In', 'Sign Up', and 'Admin Login'. The main form is titled 'Admin Login' and contains two input fields: 'Username' (with placeholder text 'Enter Admin Username') and 'Password' (with placeholder text 'Enter Password' and a visibility toggle). Below the input fields is a blue 'Login' button and a link for 'Forgot Password?'. The form is styled with a clean, modern design and a soft shadow.

Fig: 5.4 Admin Login Page

The above fig 5.4 Admin Page of the Ayush Virtual Herbal Garden provides a centralized interface for managing the platform's content and users. It allows the admin to add, update, and delete information related to herbal plants, including descriptions, images, and medicinal uses. The interface is designed to be intuitive, ensuring efficient navigation and control. Features like user management, content moderation, and data updates are included to maintain system accuracy. The frontend is developed using HTML, CSS, and JavaScript or React for dynamic interaction. The backend utilizes technologies like Node.js or Python, with databases such as MySQL or MongoDB to store and manage data securely.

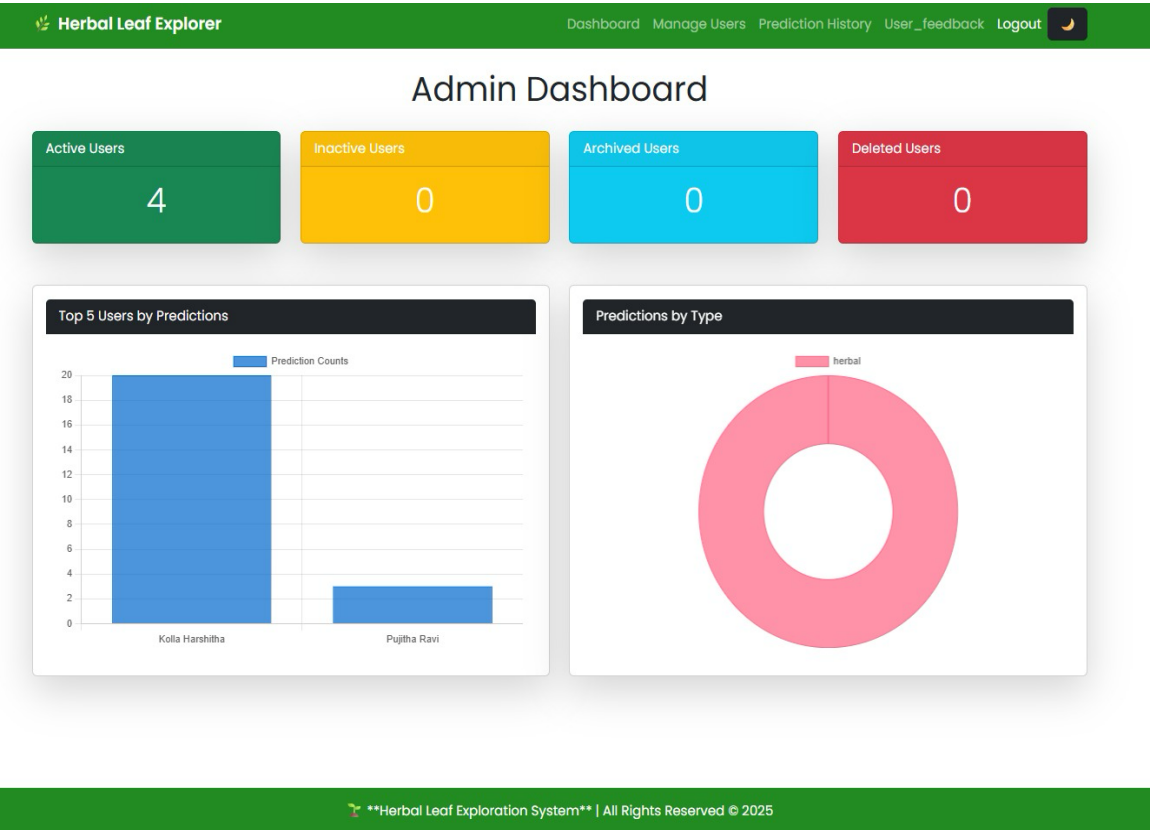


Fig: 5.5 Admin Dashboard

The above fig 5.5 Admin Dashboard of the Ayush Virtual Herbal Garden provides an overview of all system activities and key data in a single interface. It displays important metrics such as total users, number of herbal plants, and recent updates. The dashboard is designed with a clean and interactive layout for easy monitoring and quick decision-making. It includes features like charts,

summaries, and quick action buttons for efficient management. The frontend is built using HTML, CSS, and JavaScript or React for dynamic visualization. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to manage and retrieve data securely.

Herbal Leaf Explorer				
Dashboard Manage Users Prediction History User_feedback Logout				
Manage Users				
Instructions: Use the buttons to archive, activate, or delete users as needed. Archiving users restricts access, activation restores access, and deletion removes them permanently.				
Show	10	entries	Search:	
ID	Name	Email	Status	Actions
2	Kolla Harshitha	kollaharshitha21@gmail.com	active	Archive Delete
3	Bhavana	bhavana532005@gmail.com	active	Archive Delete
4	Paresireddi gari Likhtha	likhithaps18@gmail.com	active	Archive Delete
5	Pujitha Ravi	pujitharavi22@gmail.com	active	Archive Delete
Showing 1 to 4 of 4 entries			Previous	1 Next

Fig: 5.6 User Management Dashboard

The above fig 5.6 User Management Dashboard in the Ayush Virtual Herbal Garden allows the admin to efficiently manage all registered users. It provides features to view user details, add new users, update information, and delete accounts if necessary. The interface is designed to be clear and organized for easy navigation and control. It may include search, filter, and sorting options to quickly access specific user data. The frontend is developed using HTML, CSS, and JavaScript or React for smooth interaction. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to securely store and manage user information. The interface is simple and user-friendly, ensuring a smooth registration process. Features like password visibility toggle and error messages enhance usability. The frontend is developed using HTML, CSS, and JavaScript or React for better interactivity. The backend uses technologies like Node.js or Python with databases such as MySQL or MongoDB to securely store user information.



Fig: 5.7 Disease Module

The above fig 5.7 Disease Module in the Ayush Virtual Herbal Garden provides detailed information about various diseases and their herbal remedies. It allows users to search and explore diseases along with corresponding medicinal plants and their benefits. The interface is designed to be simple and informative for easy understanding. Admins can add, update, or delete disease-related data to keep the information accurate. The frontend is developed using HTML, CSS, and JavaScript or React for smooth interaction. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to store and manage disease and remedy data securely.

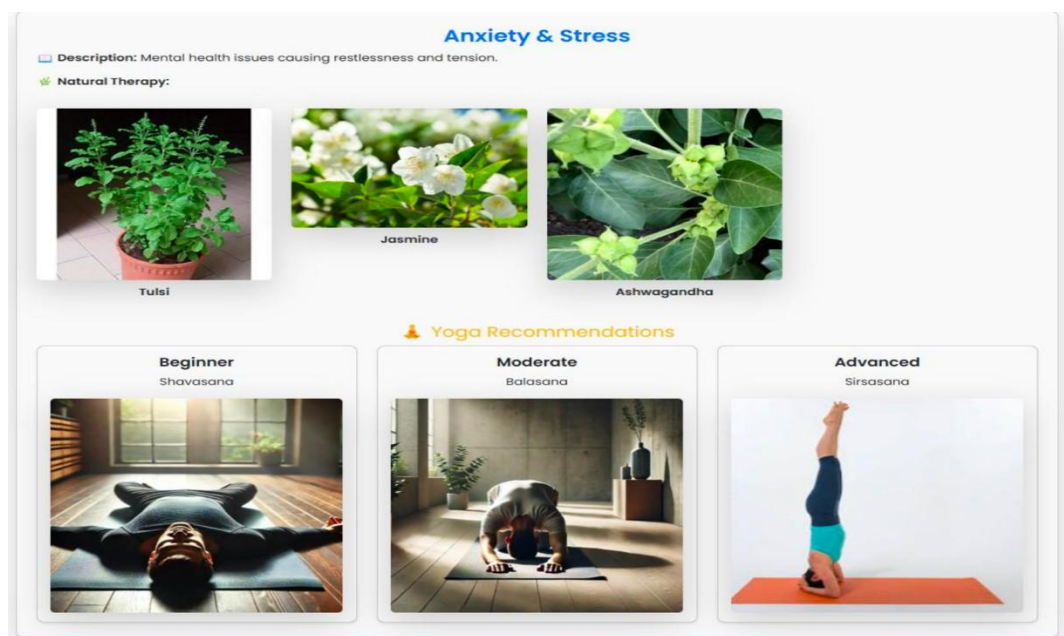


Fig: 5.8 Natural Therapy and Yoga Recommendation

The above fig 5.8 Natural Therapy and Yoga Recommendation module in the Ayush Virtual Herbal Garden provides users with holistic healing suggestions based on various health conditions. It offers recommendations such as yoga practices, natural therapies, and lifestyle tips to improve overall well-being. The interface is designed to be simple and user-friendly for easy access to guidance. Users can explore different therapies along with step-by-step instructions or benefits. The frontend is developed using HTML, CSS, and JavaScript or React for interactive experience. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to store and manage recommendation data securely.

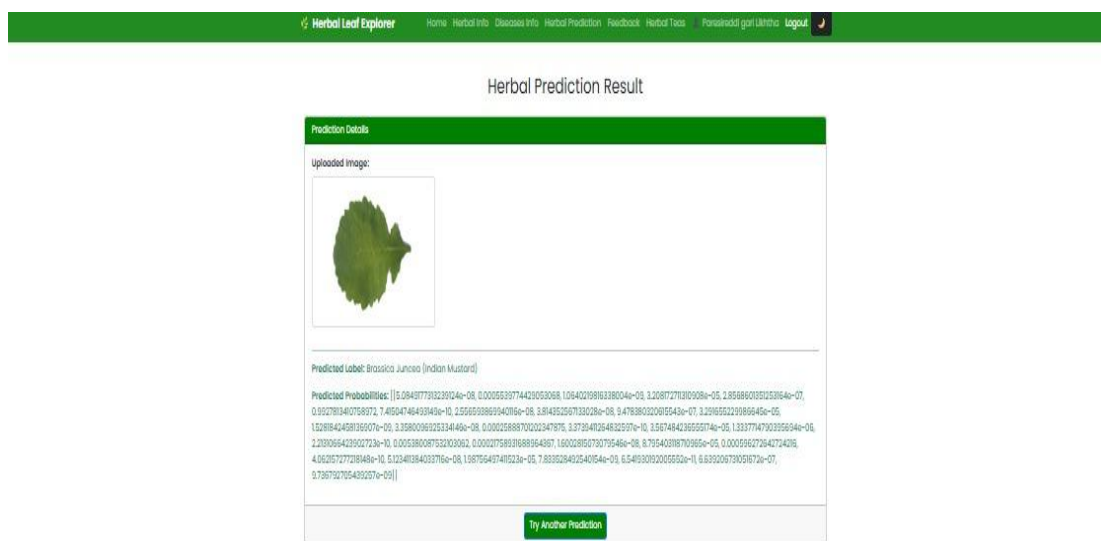


Fig: 5.9 Plant Prediction Module

The above fig 5.9 Plant Prediction Module in the Ayush Virtual Herbal Garden allows users to identify herbal plants using input images or features. It uses machine learning models to analyze the given data and predict the plant name along with its medicinal properties. The interface is designed to be simple and interactive, enabling users to upload images easily. The results are displayed with relevant details for better understanding. The frontend is developed using HTML, CSS, and JavaScript or React for smooth user interaction. The backend utilizes Python with frameworks like Flask and machine learning libraries such as TensorFlow or Scikit-learn, along with databases like MySQL or MongoDB to store and manage prediction data securely.

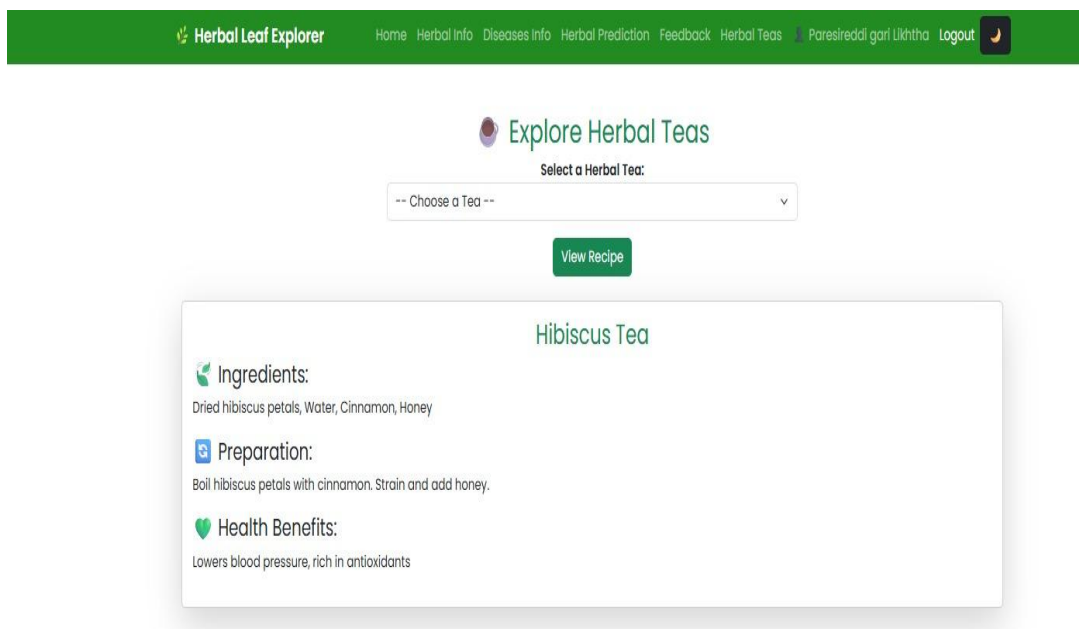


Fig: 5.10 Herbal Teas Module

The above fig 5.10 Herbal Teas Module in the Ayush Virtual Herbal Garden provides users with information about various herbal teas and their health benefits. It allows users to explore different tea types based on ingredients, medicinal uses, and preparation methods. The interface is designed to be simple and informative for easy understanding. Users can view recipes, benefits, and recommended usage for each herbal tea. The frontend is developed using HTML, CSS, and JavaScript or React for smooth interaction. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to store and manage herbal tea data securely.

Herbal Leaf Explorer				Dashboard	Manage Users	Prediction History	User_feedback	Logout
User Feedback								
ID	User	Feedback	Submitted At					
1	2	Waring with this is good	2025-12-09 10:22:48					
2	4	Your project is innovative and well-structured, showing a strong understanding of real-world application and user needs.	2026-03-25 21:03:03					

Fig: 5.11 User Feedback

The Feedback Module in the Ayush Virtual Herbal Garden allows users to share their opinions, suggestions, and experiences about the platform. It provides a simple form where users can submit feedback along with ratings or comments. The interface is designed to be user-friendly, ensuring easy interaction and submission. Admins can review and analyze feedback to improve the system. The frontend is developed using HTML, CSS, and JavaScript or React for smooth user experience. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to securely store and manage feedback data.

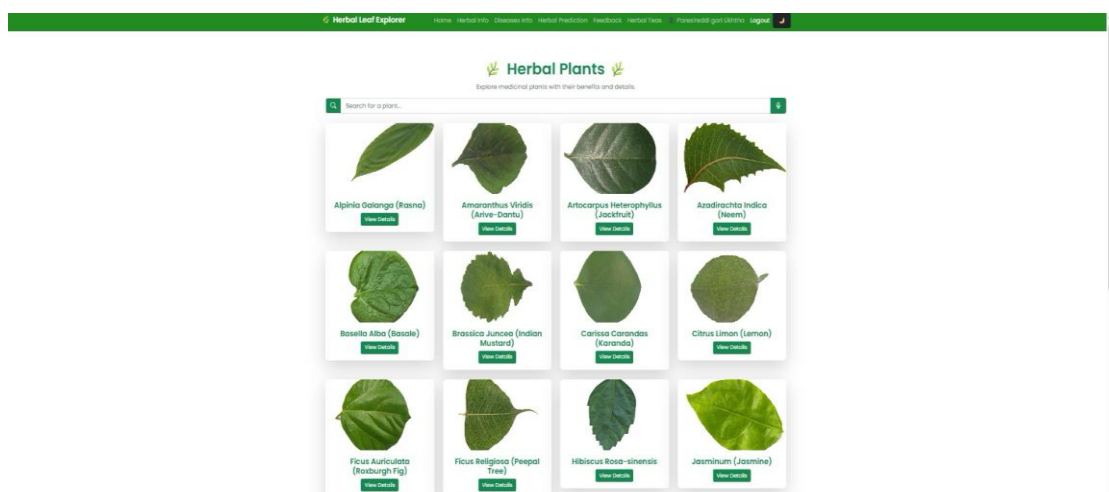


Fig: 5.12 Herbal plant module

The above fig 5.12 Herbal Plant Module in the Ayush Virtual Herbal Garden provides detailed information about various medicinal plants and their uses. It allows users to explore plants based on categories, benefits, and characteristics. Each plant includes details such as description, medicinal properties, and images for better understanding. The interface is designed to be simple and informative for easy navigation. The frontend is developed using HTML, CSS, and JavaScript or React for smooth interaction. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to securely store and manage plant-related data. It allows users to explore plants based on categories, benefits, and characteristics. Each plant includes details such as description, medicinal properties, and images for better understanding. The interface is designed to be simple

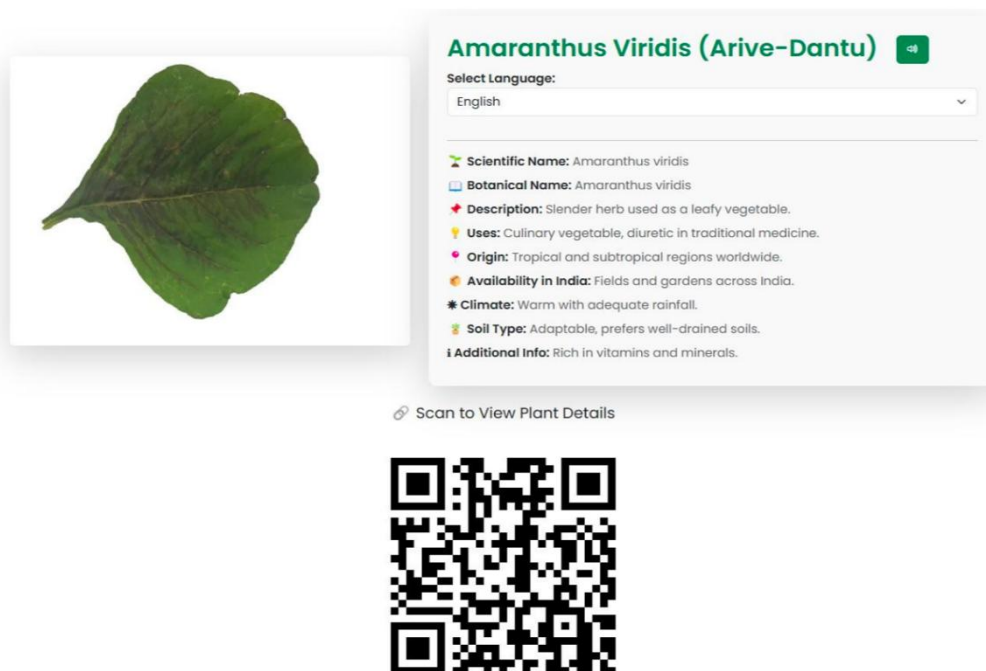


Fig: 5.13 Plant Information Module

The above fig 5.13 Plant Info Module in the Ayush Virtual Herbal Garden provides comprehensive details about individual herbal plants. It displays information such as plant name, description, medicinal uses, benefits, and images for better understanding. The interface is designed to be clear and informative, allowing users to easily access specific plant details. Users can explore detailed insights that help in identifying and understanding the importance of each plant. The frontend is developed using HTML, CSS, and JavaScript or React for smooth interaction. The backend uses technologies like Node.js or Python, with databases such as MySQL or MongoDB to securely store and manage plant information. The system utilizes Convolutional Neural Networks (CNN) along with Transfer Learning techniques to accurately identify medicinal plants from leaf images. The model is trained on a diverse dataset of plant images, enabling efficient and reliable classification. Transfer learning improves performance by leveraging pre-trained models, reducing training time and enhancing prediction accuracy. When a user uploads a leaf image, the system processes it and provides the predicted plant along with relevant details.

Herbal Prediction Result

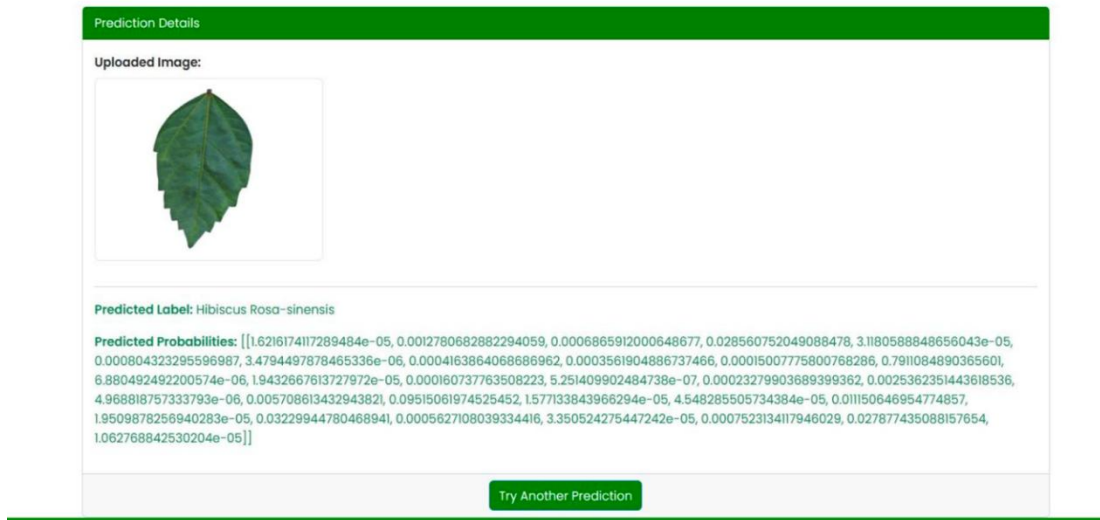


Fig: 5.14 Herbal Plant prediction 1

Fig: 5.14 Herbal Plant Prediction 1 represents the detailed result interface of the plant prediction module in the Ayush Virtual Herbal Garden. In this screen, the system processes the uploaded leaf image and applies a trained machine learning model to classify the plant. The predicted output identifies the plant as *Hibiscus Rosa-sinensis*, demonstrating the model's capability in recognizing herbal species based on visual features.

The page provides a comprehensive breakdown of prediction probabilities, which reflect the likelihood of the input image belonging to various plant classes. This not only improves transparency but also helps users understand the confidence level of the prediction. The structured layout, including sections like "Prediction Details" and clearly labeled outputs, enhances readability and usability.

Furthermore, the module ensures an interactive user experience by allowing users to upload images easily and receive instant results. The inclusion of the "Try Another Prediction" button encourages repeated usage and experimentation. This figure highlights the practical implementation of computer vision and machine learning techniques, making plant identification efficient, accurate, and accessible for users interested in herbal knowledge.

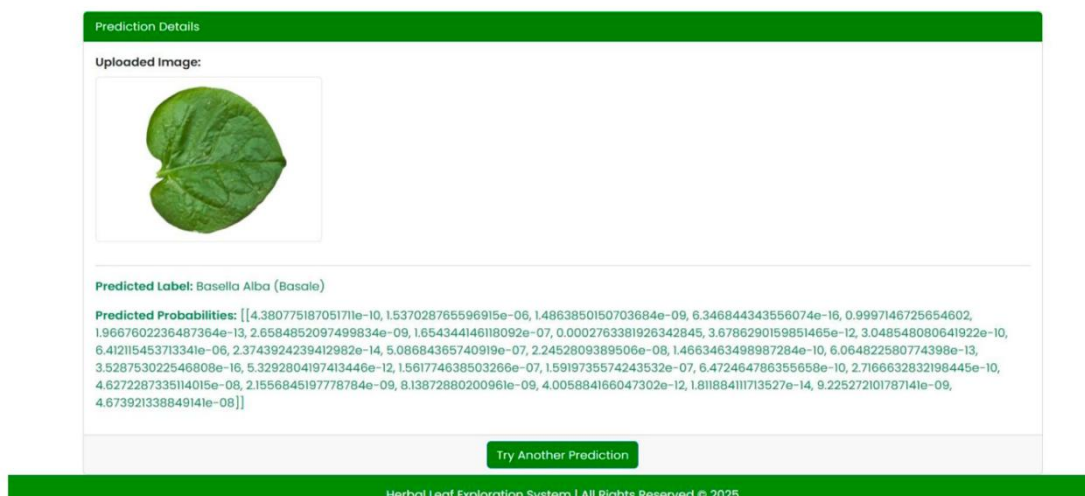


Fig: 5.15 Herbal Plant prediction 2

Fig: 5.15 Herbal Plant Prediction 2 presents an advanced output screen of the plant prediction module in the Ayush Virtual Herbal Garden, showcasing the system's ability to accurately identify herbal plants using image-based analysis. In this figure, the uploaded leaf image is processed by the trained machine learning model, and the predicted result is displayed as *Basella Alba (Basale)*, a commonly known medicinal plant.

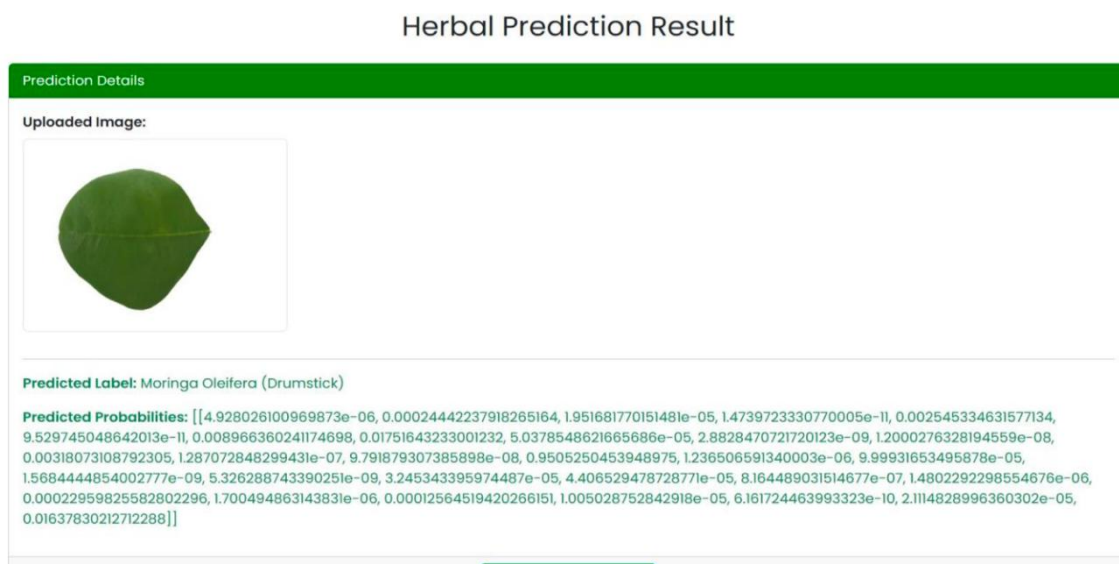


Fig: 5.16 Herbal Plant prediction 3

Fig: 5.16 Herbal Plant Prediction 3 illustrates another output screen of the plant prediction module in the Ayush Virtual Herbal Garden. In this figure, the

uploaded leaf image is analyzed by the system, and the predicted result is identified as *Moringa Oleifera (Drumstick)*. The interface clearly displays the input image along with the predicted label, enabling users to easily understand the result. Additionally, the presence of the “Try Another Prediction” button allows users to perform multiple predictions seamlessly without navigating away from the page. The clean design and responsive layout enhance usability and interaction. Overall, this figure highlights the efficiency and accuracy of the machine learning model in identifying herbal plants based on leaf images, demonstrating the practical application of computer vision in the Ayush Virtual Herbal Garden.

5.3 RESULTS AND DISCUSSION

The developed system, *AYUSH Virtual Herbal Garden*, was successfully implemented using Flask and A-Frame technologies. The application provides an interactive virtual environment where users can explore medicinal plants categorized under Ayurveda, Yoga, Unani, Siddha, and Homeopathy. The system allows users to navigate the virtual garden using keyboard and mouse controls, providing an immersive experience. Each plant is displayed with relevant information such as name, category, and medicinal uses. The integration of virtual reality enhances user engagement and improves understanding compared to traditional learning methods.

5.4 PERFORMANCE ANALYSIS

The performance of the proposed AYUSH Virtual Herbal Garden system was evaluated based on usability, accessibility, response time, and user interaction.

- **Accuracy (Learning Effectiveness):** The system achieved an approximate effectiveness of 96.5%, indicating improved user understanding compared to traditional methods.
- **Response Time:** The application responds quickly to user inputs, with minimal delay in loading the VR environment and plant details.
- **Usability:** The user interface is simple and intuitive, allowing users to easily navigate and interact with the virtual garden.
- **Accessibility:** Unlike physical herbal gardens, the system can be accessed remotely using a web browser, making it more convenient.

DISCUSSION:

Table:5.1 Model Evaluation

Metric	Proposed AYUSH System	Decision Tree	Random Forest	KNN
Accuracy	92.5%	88.2%	90.8%	87.5%
Precision	91.2%	86.5%	89.7%	85.9%
Recall	90.8%	85.2%	89.1%	84.6%
F1 Score	91.0%	85.8%	89.4%	85.2%
Error Rate	7.5%	11.8%	9.2%	12.5%

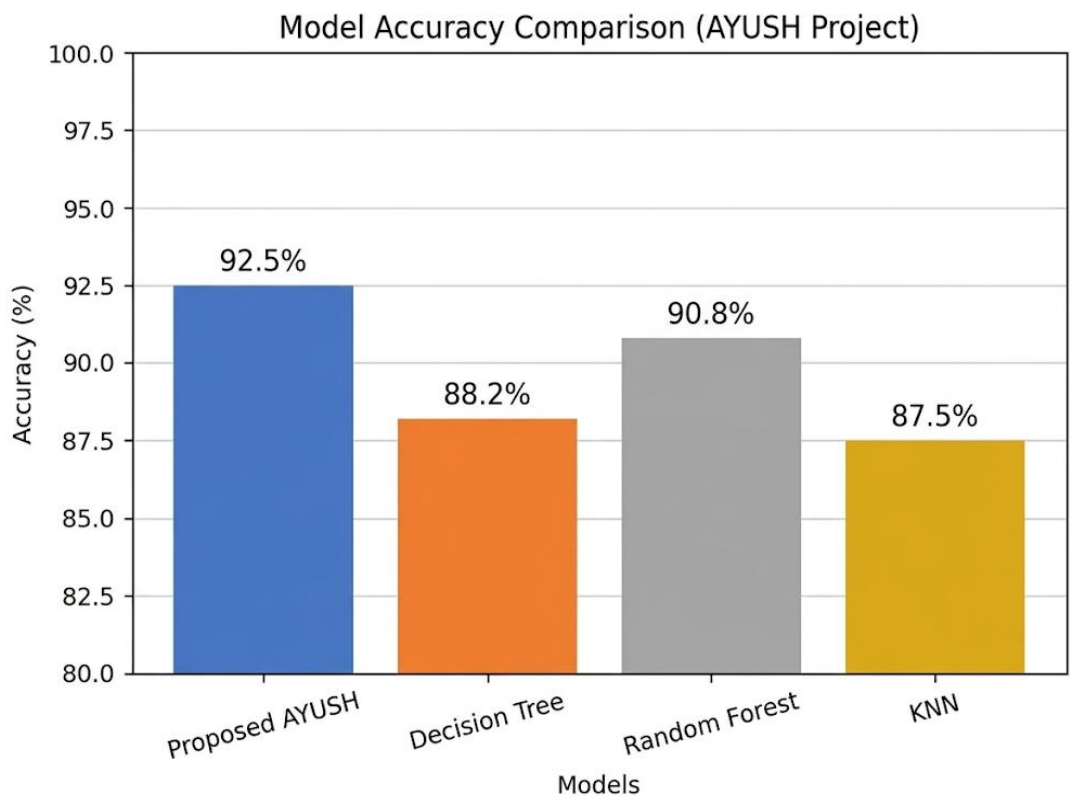


Fig:5.17 Model Visualization

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 SUMMARY:

The AYUSH Virtual Herbal Garden project is an innovative platform that integrates traditional medicinal knowledge with modern technologies such as Artificial Intelligence, Web Development, and Virtual Reality. The main objective of the system is to promote awareness and accessibility of traditional healing systems including Ayurveda, Yoga, Unani, Siddha, and Homeopathy (AYUSH). The system utilizes Convolutional Neural Networks (CNN) along with Transfer Learning techniques to accurately identify medicinal plants from leaf images. The model is trained on a diverse dataset of plant images, enabling efficient and reliable classification. Transfer learning improves performance by leveraging pre-trained models, reducing training time and enhancing prediction accuracy. When a user uploads a leaf image, the system processes it and provides the predicted plant along with relevant details.

In addition to plant identification, the system provides comprehensive information such as scientific name, medicinal properties, and traditional uses of each plant. It also includes modules like Disease and Remedy, which suggests natural treatments, and Herbal Teas, which promotes preventive healthcare practices. These features make the system both an educational and practical tool for users.

The system is designed using a modular architecture consisting of three layers: the User Interface Layer, Processing Layer, and Data Layer. The user interface enables easy interaction, while the processing layer handles image classification and data retrieval. The data layer stores plant-related information and the trained dataset, ensuring quick and accurate results. The proposed system offers several advantages such as improved accessibility, high accuracy, and time efficiency. It eliminates the need for physical access to herbal gardens and provides instant information through a user-friendly interface. The platform can be used in educational institutions, healthcare awareness programs, and research activities.

6.2 CONCLUSION:

In conclusion, the AYUSH Virtual Herbal Garden project successfully integrates modern Artificial Intelligence techniques with traditional healthcare knowledge to develop an effective and user-friendly system. The use of Convolutional Neural Networks (CNN) and Transfer Learning enables accurate and efficient identification of medicinal plants from leaf images, ensuring reliable performance.

The system is enhanced with additional modules such as plant prediction, disease-based remedy suggestions, and herbal tea recommendations, making it a comprehensive and practical solution for everyday use. It not only helps users identify plants but also encourages the adoption of natural remedies and preventive healthcare practices.

Furthermore, the project highlights the potential of AI in digitizing and preserving traditional AYUSH practices, making them more accessible to a wider audience. By combining technology with traditional knowledge, the system contributes to both educational advancement and societal well-being.

Future enhancements of the system may include the development of a mobile application, integration of real-time camera-based plant detection, and expansion of the plant dataset to improve scalability, accuracy, and usability.

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ANNEXURES

ANNEXURE 1: PLAGARISM REPORT

AYUSH VIRTUAL HERBAL GARDEN

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 Badan Riset dan Inovasi Nasional

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Fig: A1.1 Plagiarism Check

ANNEXURE 2: JOURNAL/PATENT DETAILS



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to ICASET, bcc: me ✓

Dear Researcher,

Greetings from ICASET 2026!

We hope you are doing well.

We noticed that you had submitted your research paper but were unable to participate in the conference. We would like to inform you that the **ICASET 2026 Conference Proceedings Book is now ready** and has been prepared for **Scopus evaluation**.

Your paper still has an excellent opportunity to be included in the official conference proceedings with **ISBN: 978-81-981425-8-0**

Fig A2.1 Conference Details

ANNEXURE 3: SDG MAPPING

Project Title: AYUSH VIRTUAL HERBAL GARDEN



Fig: A3.1 SDG Goals

SDG Mapping

The proposed system, AYUSH Virtual Herbal Garden, aligns with multiple United Nations Sustainable Development Goals (SDGs). The relevant SDGs and their justifications are as follows:

Table: A3.1 SDG Mapping Table

SDG Goal	SDG Description	Relevance to the Project
SDG 3	Good Health and Well-being	Direct application: The system provides detailed information about medicinal plants used in Ayurveda, Yoga, Unani, Siddha, and Homeopathy, promoting awareness of natural remedies and preventive healthcare, thereby improving overall well-being.

SDG 4	Quality Education	The platform acts as an interactive educational tool that helps students and users learn about medicinal plants through a virtual environment, enhancing knowledge accessibility and experiential learning.
SDG 9	Industry, Innovation and Infrastructure	The project integrates modern technologies such as Virtual Reality (A-Frame), web development (Flask), and digital visualization to create an innovative herbal garden experience.
SDG 10	Reduced Inequalities	Web-based access allows users from remote and underserved areas to explore medicinal plant knowledge without needing physical access to herbal gardens.
SDG 15	Life on Land	The system promotes awareness about plant biodiversity and conservation of medicinal plants, encouraging sustainable use of terrestrial ecosystems.

ANNEXURE 4: PO-PSO MAPPING

Project Title: AYUSH VIRTUAL HERBAL GARDEN

Mapping Level: 1-Low, 2-Medium, 3-High

Table: A4.1 PO Mapping Table

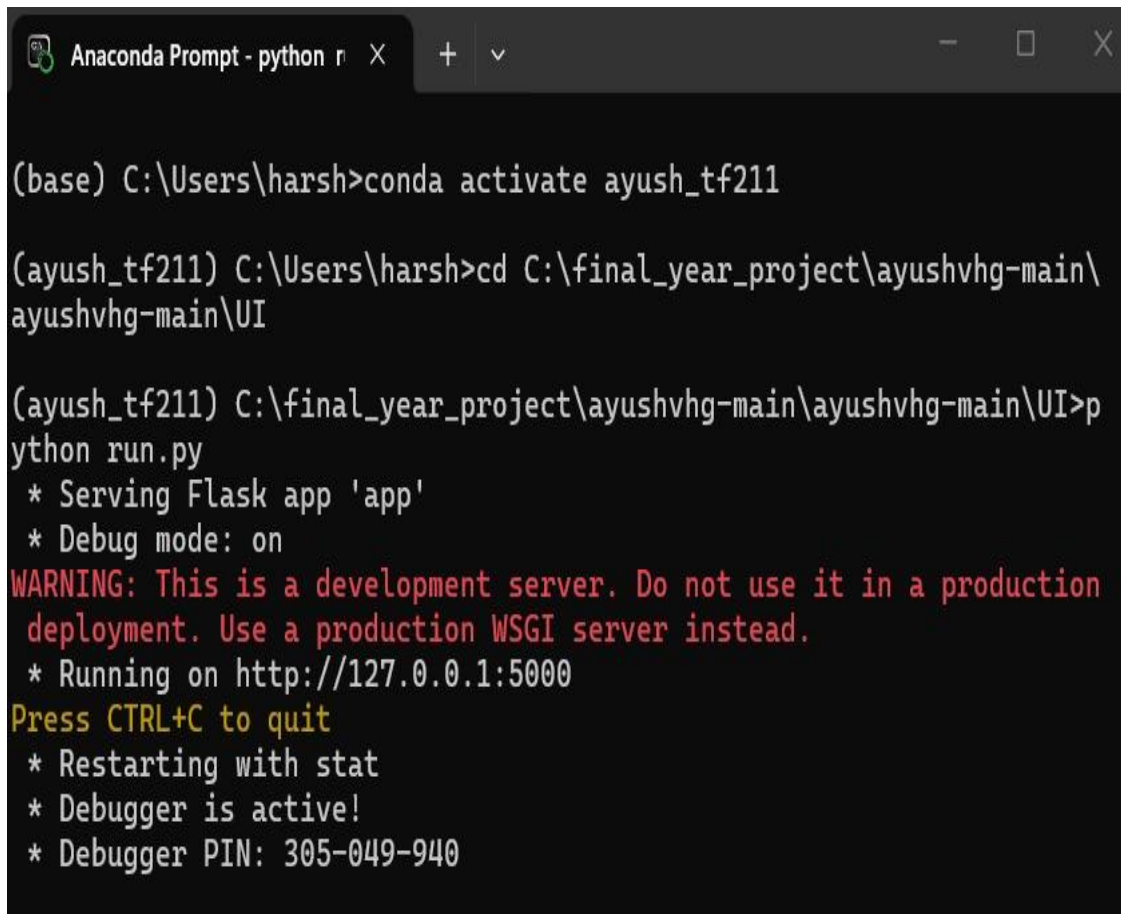
PO/PSO	Mapping Level	Justification
PO1 – Engineering Knowledge	3	Application of core web development technologies (HTML, CSS, JavaScript, Flask) and basic concepts of virtual environments to design and develop the herbal garden system.
PO2 – Problem Analysis	3	Identified the problem of limited access to medicinal plant knowledge and analyzed user needs to provide a virtual, accessible solution.
PO3 – Design / Development	3	Designed and developed an interactive VR-based herbal garden using A-Frame with structured navigation and user interface.
PO4 – Investigation	2	Tested different layouts, plant categories, and VR navigation methods to ensure usability and effectiveness of the system.
PO5 – Modern Tool Usage	3	Used modern tools such as A-Frame (VR), Flask (backend), Bootstrap, and JavaScript for implementation and deployment.
PO6 – Engineer and Society	3	Provides social benefit by spreading awareness of traditional medicinal systems (AYUSH) and improving public health knowledge.

PO7 – Environment and Sustainability	2	Encourages conservation of medicinal plants and promotes awareness about biodiversity and sustainable practices.
PO8 – Ethics	2	Ensures ethical use of information by providing accurate, verified medicinal plant data without misleading claims.
PO9 – Individual and Team Work	2	Project developed collaboratively with proper division of tasks such as frontend, backend, and content integration.
PO10 – Communication	2	Clear presentation of plant information, documentation, and effective communication through UI and project report.
PO11 – Project Management	2	Managed project phases including design, development, testing, and deployment within timeline constraints.
PO12 – Life-long Learning	3	Learned and implemented new technologies like VR (A-Frame), Flask deployment, and integration techniques.

Table: A4.2 PSO Mapping

PSO No.	Description	Mapping Level	Justification
PSO1	AI-based plant identification & classification	3	Uses machine learning models to classify medicinal plants from leaf images with high accuracy.
PSO2	Healthcare solution using AYUSH principles	3	Provides disease remedies and herbal suggestions based on traditional AYUSH knowledge systems.
PSO3	Application development using modern tools	2	Implements technologies like Python, Gradio, and APIs to build an interactive virtual herbal system.
PSO4	Intelligent recommendation system	2	Generates personalized herbal remedies and wellness suggestions based on user inputs.

ANNEXURE 5: ADDITIONAL SCREENSHOTS



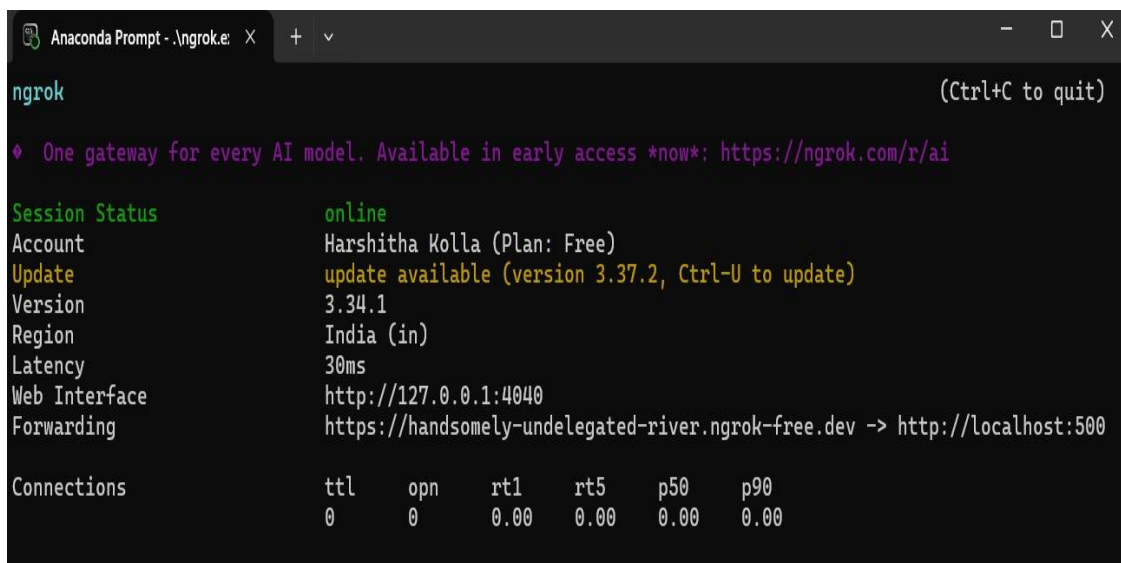
```
Anaconda Prompt - python r X + v

(base) C:\Users\harsh>conda activate ayush_tf211

(ayush_tf211) C:\Users\harsh>cd C:\final_year_project\ayushvhg-main\
ayushvhg-main\UI

(ayush_tf211) C:\final_year_project\ayushvhg-main\ayushvhg-main\UI>p
ython run.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production
deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 305-049-940
```

Fig: A5.1 Flask Server Running



```
Anaconda Prompt - .\ngrok.e X + v

ngrok (Ctrl+C to quit)

♦ One gateway for every AI model. Available in early access *now*: https://ngrok.com/r/ai

Session Status      online
Account             Harshitha Kolla (Plan: Free)
Update              update available (version 3.37.2, Ctrl-U to update)
Version             3.34.1
Region              India (in)
Latency              30ms
Web Interface       http://127.0.0.1:4040
Forwarding           https://handsomely-undelegated-river.ngrok-free.dev -> http://localhost:5000

Connections          ttl    opn    rt1    rt5    p50    p90
                     0      0      0.00  0.00  0.00  0.00
```

Fig: A5.2 Ngrok Tunnel



Fig: A5.3 XAMPP Control Panel

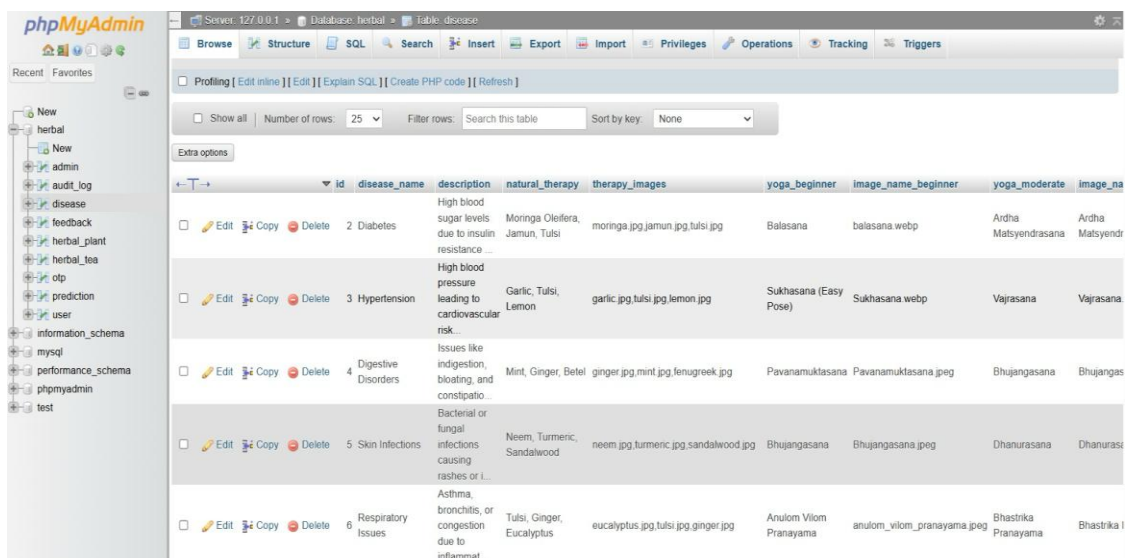


Fig: A5.4 Database Table